
SYMBOLIC REGRESSION CAN PERFORM ON PAR WITH TREE-BASED ENSEMBLES IN REGRESSION PROBLEMS: SUPPLEMENTARY MATERIALS

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ABSTRACT

This document is intended as supplementary materials for the paper *Tonda A. and Lutton E. 2026. Symbolic Regression Can Perform on Par with Tree-based Ensembles in Regression Problems, Proceedings of the Biennial International Conference on Artificial Evolution (EA) 2026*. You can find the full paper here: <https://albertotonda.github.io/publication/tonda-2026-symbolic/tonda-2026-symbolic.pdf>.

Keywords Benchmarking · Extreme Gradient Boosting · Random Forest · Regression · Symbolic Regression

A A note on the Supplementary Materials

The sections of this document are not meant to be a proper part of a scientific paper, yet; they are added to the main text to provide extra information, should the reader be interested in more detailed explanations. The text is still in a draft format, and it will need to be reworked before being considered a scientific document. In particular, equations created by Symbolic Regression (SR) are often extremely large and sometimes they don't fit inside the page, even when using solutions such as reducing text size or rotating the page. The text of the main paper already contains all the necessary information to evaluate the proposed approach.

B Considerations on the datasets

B.1 white_wine and red_wine

Inspecting the datasets for the two tasks 361249 (white_wine) and 361250 (red_wine), on which SR performs significantly worse than the other approaches using both default and enhanced hyperparameters, reveals them to be quite unusual among regression problems. Both datasets present quantitative, real-valued features about wine (such as pH, density, content of sulfur dioxide, etc.), with the target y being an assessment of quality performed by human experts. This quality assessment is represented by an integer, which in dataset white_wine ranges from 3 to 9, for a total of only seven different values among the 4,898 samples, and in red_wine, the quality label ranges from 3 to 8, for just six different values for 1,599 samples.

It is interesting to notice that the same tasks could be easily reframed as classification problems, as the target $y \in \mathbb{N}^+$; and in fact, the Machine Learning (ML) domain defines problems of this kind as *ordinal classification* or *ordinal regression* [3], because the relative values of the integer labels have a meaning (e.g., in the case of these datasets, an evaluation of 5 is better than an evaluation of 4). Random Forest (RF) and eXtreme Gradient Boosting (XGBoost) employ similar strategies for both regression and classification, thus it makes sense that they might be particularly suited for problems that lie at the edge between the two tasks.

Using a classical MSE-based fitness for ordinal regression might not be the best approach; unfortunately, after a cursory exploration of specialized literature, we could not find any SR fitness function specifically tailored for ordinal regression. We thus decided to experiment with a variation of Poisson likelihood:

$$\mathcal{L}(\hat{y}, y) = \hat{y} - y + y \cdot \ln\left(\frac{y}{\hat{y}}\right) \quad (1)$$

Where $\hat{y}$ is the predicted value and y is the actual label. To avoid numerical issues, in practice we will use $\hat{y}' = \max(\hat{y}, \epsilon)$ instead of $\hat{y}$, with $\epsilon = 10^{-8}$. Still, no improvement has been observed so far.

B.2 naval_propulsion_plant

The dataset for task 361247 reveals a situation somehow similar to the one described in subsection B.1; while `naval_propulsion_plant` has 11,934 samples, the target y only features 51 distinct values, with $y \in [0.95, 1.00]$ with increments of 0.001. A comparatively small amount of different values for the target can give an edge to approaches based on Decision Trees (DTs), which partition the feature space into a set of regions and predict a constant value within each region. It is also worth mentioning that 361247 is the only task for which PySR with default hyperparameters (less computational resources, narrower search space) performs significantly *better* than the same algorithm with enhanced hyperparameters. We hypothesize that this is probably due to a premature stop of the evolutionary search: In this particular case, the larger search space (more operators, higher maximum complexity) might be more difficult to explore, especially if SR quickly started considering more complex equations, which are typically slower to evaluate.

In order to test this hypothesis, we rerun the algorithm, increasing the timeout to 5 hours, and the mean performance indeed improved to $R^2 = 0.7108 \pm 0.1223$.

B.3 superconductivity

Another dataset with a relatively small number of unique values for the target when compared to the number of samples: 3,007 distinct values over 21,263 samples.

B.4 pumadyn32nh

On the OpenML website, the description of the `pumadyn32nh` dataset used in task 361259¹ is: “A realistic simulation of the dynamics of a Puma 560 robot arm. The task in this dataset is to predict the angular acceleration of one of the robot arm’s links. The task is nonlinear and has high noise. The simulator (sic) got as input the angular positions, velocities and torques and other dynamic parameters of the Puma arm. The goal was to predict the acceleration of link 6 *thetad6*”.

This is one of the datasets where SR performs better than the tree-based ensembles, and observing the quite compact equation obtained through the Large Language Model (LLM)-assisted analysis in subsection E.13, $\hat{y} = c_0 x_{15} \cos(x_4)$, we can hypothesize that it’s probably due to the availability of the transcendental functions among the SR operators, as modeling angular acceleration clearly benefits from functions typically working on angles.

B.5 fifa

The equation found for `fifa` employs a single feature, x_4 , which corresponds to the feature called `overall`. While the dataset description found on the OpenML website does not explicitly describe all features in detail, searching on the internet revealed that `overall` corresponds to the best overall ranking of the player. It thus makes sense that the wage of a player is mostly, if not entirely, explained by their relative ranking.

C Search space of hyperparameter optimization

In the following, we report the (min, max) range or the list of options [option1, option2] used during hyperparameter optimization for RF and XGBoost with Optuna [1].

C.1 Random Forest

¹<http://openml.org/search?type=data&status=active&id=44981>

```

1  "n_estimators": {
2      "min": 100,
3      "max": 5000
4  },
5  "max_depth": {
6      "min": 3,
7      "max": 30
8  },
9  "min_samples_split": {
10     "min": 2,
11     "max": 20
12 },
13 "min_samples_leaf": {
14     "min": 1,
15     "max": 20
16 },
17 "max_features": {
18     "min": 0.1,
19     "max": 1.0
20 },
21 "n_jobs": -1,
22 "random_state": 42
23 },

```

C.2 eXtreme Gradient Boosting

```

1  "n_estimators": {
2      "min": 100,
3      "max": 5000
4  },
5  "max_depth": {
6      "min": 3,
7      "max": 30
8  },
9  "min_leaves": {
10     "min": 0,
11     "max": 64
12 },
13 "learning_rate": {
14     "min": 0.001,
15     "max": 0.3
16 },
17 "subsample": {
18     "min": 0.5,
19     "max": 1.0
20 },
21 "colsample_bytree": {
22     "min": 0.5,
23     "max": 1.0
24 },
25 "colsample_bylevel": {
26     "min": 0.5,
27     "max": 1.0
28 },
29 "min_child_weight": {
30     "min": 1,
31     "max": 20
32 },
33 "gamma": {
34     "min": 1e-08,
35     "max": 10.0
36 },
37 "reg_alpha": {

```

```

38     "min": 1e-08,
39     "max": 10.0
40 },
41 "reg_lambda": {
42     "min": 1e-08,
43     "max": 10.0
44 },
45 "booster": [
46     "gbtree",
47     "dart"
48 ],
49 "grow_policy": [
50     "depthwise",
51     "lossguide"
52 ],
53 "bootstrap": true,
54 "n_jobs": -1,
55 "random_state": 42

```

D Enhanced hyperparameters for PySR

```

1  "niterations": 1000,
2  "population_size": 100,
3  "maxsize": 300,
4  "binary_operators": [
5      "+",
6      "-",
7      "*",
8      "/"
9  ],
10 "unary_operators": [
11     "sin",
12     "cos",
13     "log",
14     "exp"
15 ],
16 "temp_equation_file": true,
17 "random_state": 42,
18 "procs": null,
19 "parallelism": "multiprocessing",
20 "timeout_in_seconds": 1440

```

E LLM-assisted analysis of SR equations

One of the stated advantages of Symbolic Regression over other types of regressors is the alleged interpretability of the equations. However, the experimental evaluation showed that in order to become competitive with state-of-the-art tree-based models, it is necessary to increase the maximum allowed equation size, add extra operators to the building blocks, and use a considerable extra amount of computational resources. To assess possible loss of interpretability, in this section we will qualitatively analyze selected equations discovered by Symbolic Regression. As we aim to perform this analysis for multiple datasets, we choose to employ the support of a Large Language Model (LLM), the free version of Google Gemini® 3.1, accessed between May 19 and June 6, 2026, using the “Pro” mode with the “Extended reasoning” option. After uploading the $\LaTeX$ equation files to Gemini, we employ the following prompt:

First prompt for the LLM-assisted analysis

Hello Gemini! I am going to pass you several files containing equations in $\LaTeX$, can you tell me what are the similarities? For example, similarities in structure (same terms) or presence of the same variables (numbered $x_1, \dots, x_n$). I am interested in the common parts of all these equations.

In case the model did not provide a generic equation summarizing all the candidate solution after the first prompt, a second prompt with an explicit request was used:

Second prompt for the LLM-assisted analysis

Going back to the equations, could you create a generic expression catching most of the common parts with undefined constants such as $c_1, c_2, \dots$?

The results of the automatic analysis performed by Gemini are then manually inspected for errors, and used as a base to further develop an expert analysis.

The datasets analyzed in the following were selected according to the following criteria:

1. Either SR performed on par with or significantly better than the other approaches;
2. Or SR's performance achieved $R^2 > 0.6$.

E.1 Task 361616, default hyperparameters

In task 361616 (Moneyball), Symbolic Regression already outperforms the tree-based ensembles even with default hyperparameters. The equations selected using the validation set method for each fold are:

$$\hat{y} = x_4 \cdot 2.73 + (x_3 + x_6 \cdot 3.67 \cdot 10^3) (x_5 - 0.0473) - (x_9 \cdot (x_1 - x_9 + 2.30) + 125.) + 5.72 \quad (2)$$

$$\hat{y} = x_4 + x_5 \left(x_3 + \frac{x_6 (x_4 + x_9 + 103.)}{0.0611} \right) - x_8 - \left(\frac{x_1}{x_7} - x_6 x_9 \right) \quad (3)$$

$$\hat{y} = x_2 (-0.432) x_6 - x_3 (-0.261) + (x_4 + x_5 (x_6 \cdot 2.47 \cdot 10^3 - 172.)) 2.52 \quad (4)$$

$$\hat{y} = -0.00388 x_2 x_4 + x_4 \cdot 10.7 + x_5 (x_3 + x_6 \cdot 3.28 \cdot 10^3) - 188. \quad (5)$$

$$\hat{y} = \left(x_4 + \left(-(-2.03) x_3 + x_6 \left(1.86 \cdot 10^4 - \frac{3.47 \cdot 10^3}{x_5} \right) \right) 0.0511 - 60.2 \right) 2.79 \quad (6)$$

$$\hat{y} = (x_3 + x_6 \cdot 4.00 \cdot 10^3) 0.846 x_5 - (146. + x_4 (-2.76)) \quad (7)$$

$$\hat{y} = -\frac{x_1}{0.242} + x_2 (-0.286) + x_5 (x_3 \cdot 0.874 + x_6 \cdot 3.46 \cdot 10^3) - (x_4 + x_9) (-2.71) + 401. \quad (8)$$

$$\hat{y} = -x_1 - x_8 - x_9 (x_1 + x_9 (-0.809)) + (x_3 + x_4 \cdot 9.76 + 577.) (x_5 + x_6 \cdot 0.502 - 0.251) + 128. \quad (9)$$

$$\hat{y} = -x_1 - x_1 - x_1 + x_4 + x_4 + x_5 \cdot 1.92 \cdot 10^3 + x_6 x_6 \cdot 1.31 \cdot 10^3 - (x_3 (-0.310) - x_4 - 1.07) - 584. \quad (10)$$

$$\hat{y} = -x_1 + x_5 (x_5 (-0.955) - x_6 \cdot 2.92) \left(x_2 \cdot 137. \frac{1}{x_3} - 1.55 \cdot 10^3 \right) + (-x_1 + x_4 + x_9) 2.56 - 63.7 \quad (11)$$

The equations are already quite intricate, even with the initial limitation of complexity to 30, and finding common elements without automatic support would require a considerable amount of manual inspection. The text produced by the LLM in response to the prompt in Text Box E claims to have found the following patterns:

1. Variables x_3, x_4, x_5 , and x_6 appear in every equation;
2. The majority of equations feature a linear additive term as a function of x_4 , with a constant $c_1 \approx 2.7$;
3. A pattern $x_5 \cdot (x_3 + c_2 \cdot x_6)$ also appears frequently, with c_2 typically very large $c_2 \approx 10^3$.

After concluding the analysis, the LLM suggests the following minimal equation form:

$$\hat{y} \approx c_1 x_4 + x_5 (x_3 + c_2 x_6) \quad (12)$$

We test the proposed equation 12, optimizing its constants c_1, c_2 on the whole Moneyball dataset using the Nelder-Mead algorithm. The results, however, are not satisfying, with a negative R^2 value. Looking back at the analysis, the LLM did not consider that x_3 could also be multiplied by a constant, and that all equations 2-11 contain a separate constant term. Reformulating the equation as:

$$\hat{y} \approx c_1 x_4 + x_5 (c_2 x_3 + c_3 x_6) + c_4 \quad (13)$$

And re-optimizing constants $c_1, \dots, c_4$ again using Nelder-Mead on the whole dataset, this time we obtain $R^2 = 0.9458$ with the final equation being:

$$\hat{y} = 2.776x_4 + x_5 (0.860x_3 + 3380.399x_6) - 150.374, \quad R^2 = 0.9458 \quad (14)$$

The performance of this equation is on par with the mean $R^2 = 0.9453$, and even if it appears slightly higher, it is important to consider that the constants were optimized on the whole dataset, without a cross-validation, so overfitting might play a small role. As a final validation, we perform a 10-fold cross-validation using the task’s default folds, optimizing the constants on the training set and evaluating the R^2 value on the test, for a final mean $R^2 = 0.9454 \pm 0.0048$.

E.2 Task 361234, default hyperparameters

Task 361234 (aba1one) is the other example where Symbolic Regression outperforms tree-based ensembles even with default hyperparameters. Here below are the equations found for each fold:

$$\hat{y} = \left(-0.202 + \frac{x_4 + x_7 + 0.0828}{x_5 + 0.0721} \right) \left(-x_6 + 2.88 + \frac{x_3 + x_4 - x_6}{x_1} \right) \quad (15)$$

$$\hat{y} = x_3 + x_3 + 2.48 + \frac{-x_4 - x_4 (-0.506) + x_6 - 1.44}{(-1) 0.103 + \frac{x_4 + \frac{x_5 + 0.0633}{-0.297}}{20.9x_7}} \quad (16)$$

$$\hat{y} = \frac{x_4 (2.34 - x_6) \left(x_2 + x_3 + 0.283 - \frac{0.0357}{x_2} + \frac{x_7}{x_1} \right)}{0.742x_5} + 2.09 \quad (17)$$

$$\hat{y} = -(x_1 - x_5x_4 \cdot 0.779 + x_6 \cdot 7.33) + 3.43 + \frac{\frac{1.87 - x_3}{-x_7 - 0.271} + 7.09}{\frac{1}{x_4}x_5} \quad (18)$$

$$\hat{y} = -x_6 + \left(x_4 + x_7 \cdot 5.04 \frac{1}{x_5 + 0.174} - \left(x_6 - \left(-x_6 + \frac{x_4 - x_7}{x_5} \right) \right) 1.07 - 0.325 \right) 2.49 + 0.129 \quad (19)$$

$$\hat{y} = x_6 + (x_4 - x_6) \left(-(x_7 + x_7) + \frac{(x_3 + x_7) 4.38 + 3.65 + \frac{x_6}{x_1}}{x_5 - -0.0228} \right) + 0.709 \quad (20)$$

$$\hat{y} = -x_1 + x_4 - (x_1 - 0.513) + \left(x_3 + \frac{x_4 + x_7 - (x_5 + x_6)}{x_5 + 0.143} \right) 5.90 + 3.01 \quad (21)$$

$$\hat{y} = (-1) (-1.06) + \frac{-x_4 - 1.88}{\frac{1}{-10.3} \left(-x_3 + x_5 + \frac{x_5}{x_7} + x_6 + \frac{x_1 + x_5}{x_4} \right)} \quad (22)$$

$$\hat{y} = -x_2 + x_4 + x_5 + x_7 \cdot 8.41 \frac{1}{x_5 + (x_4 - x_6 + x_7 (-1.30)) (-0.366) + 0.0676} + 2.84 \quad (23)$$

$$\hat{y} = -6.75 + \frac{4.11 (x_3 - (x_6 + 1.17) + (x_4 + x_7 - -0.996) 1.63)}{x_5 + 0.202} + \frac{x_5 + x_5}{x_1} \quad (24)$$

The common patterns alleged by the LLM-generated text in response to the prompt in E are:

1. The presence of x_5 in the denominator in each equation;
2. Every equation is built around a single main fractional term multiplied by a constant;
3. The variables located in the numerators (x_4 , x_2 , or x_7) are strictly linear; no variables are raised to a power;
4. Variable frequency: x_5 (10/10), x_4 (7/10), x_7 (3/10), x_2 (2/10).

Furthermore, the text states that the equations could be grouped in three different clusters, featuring canonical forms:

$$\hat{y} \approx c_1 \frac{x_4}{x_5 + c_2} \quad (25)$$

$$\hat{y} \approx c_1 + c_2 \frac{x_7}{x_5 + c_3} \quad (26)$$

$$\hat{y} \approx c_1 \frac{x_2 x_4}{x_5} \quad (27)$$

After a manual inspection, we realize that the LLM text ignores the presence of x_6 (in 10/10 equations!), x_3 (8/10), and x_1 (6/10). Starting from equations 25-27, but also considering x_6 and x_3 we design a new equation:

$$\hat{y} \approx \frac{c_1x_3 + c_2x_4 + c_3x_6 + c_4}{c_5x_5 + c_6x_7 + c_7} + c_8 \quad (28)$$

This time, optimizing constants $c_1, \dots, c_8$ with Nelder-Mead on the whole `abalone` dataset gives inconsistent results, so we switch to CMA-ES [2] for evolutionary numerical optimization. The final best result has a $R^2 = 0.5667$, comparable to the original $R^2 = 0.5652$, and higher than XGBoost’s at $R^2 = 0.4919$, with final equation:

$$\hat{y} = -3.259 + \frac{19.441x_3 + 44.094x_4 - 34.611x_6 + 7.709}{5.870x_5 - 1.487x_7 + 1.236}, \quad R^2 = 0.5667 \quad (29)$$

The final R^2 value might appear slightly higher, but it is important to notice that for this qualitative study the constants of Eq. 28 have been optimized on the whole dataset, so overfitting might play a role. As a final validation, we perform a 10-fold cross-validation using the task’s default folds, optimizing the constants on the training set and evaluating the R^2 value on the test, for a final mean $R^2 = 0.5662 \pm 0.0254$.

E.3 Task 361622, enhanced hyperparameters

In task 361622 (`cars`), Symbolic Regression is already non-significantly different from Random Forest and XGBoost with default hyperparameters; however, with enhanced hyperparameters its mean R^2 value improves significantly, so in the following we report the results for the latter case. As the maximum size for equations is set to 300 and more operands can be used, we expect considerably more complex solutions than in cases with default hyperparameters. The equations selected using the validation set method for each fold are:

$$\hat{y} = -(x_0 \cdot 0.164 + (x_1 - x_7) e^{\cos(x_{15}x_3) + \frac{3.67}{x_2} x_1 \cdot 9.46 + 1.06}) + \left(x_1 \cdot 3.08 \cdot 10^3 + (x_1 - x_{14}) \left(-(-6.97 \cdot 10^3) x_{10} - x_7 \left(-4.11 \cdot 10^3 \right) + \cos \left(\frac{x_{13}}{x_9 - 1.51} \right) (x_1 - \sin(-(x_1 - x_{10}) + e^{x_{11}})) (-195.) + e^{x_1 \left(\cos(x_{15}) + \frac{e^{x_2}}{0.192x_0 \sin(x_1)} \right)} + 4.41 \cdot 10^3 \right) + (x_1 x_{12} + e^{\sin(x_{16})}) \cdot 6.66 \cdot 10^3 \right) + e^{x_6 + 7.03} \quad (30)$$

$$\hat{y} = \left((1.14x_1 + 3.88x_{13} + 3.88x_7 + 4.08 \sin(x_1x_6 + x_1 + x_{14} - x_{15} - 1.78)) (x_2 + e^{x_{10}} + \sin(\log(x_0))) - \cos(x_{15} - 2.02) \right) + (x_{11} - x_{15} + \cos(x_1x_9 + x_1 + (-3.69x_6 + x_8 - 6.79) e^{x_{15} \log(x_0)} + \sin((\cos(3.69x_6x_{15}) + x_1 - (\log(x_0) + 21.7) (-x_{14} + \sin(x_0) + 7.44) e^{(x_{13} - x_5 + x_8) e^{-\sin(-x_1 + x_{12} + x_8 + 0.841)})) \quad (31)$$

$$\hat{y} = -\frac{x_0}{-x_{12} - x_7 + 5.54} + x_1(x_{13} - 1.00x_8) \left(2x_1^2x_5(x_1 - 11.8x_9) + 9.43 \right) - 2x_{13} - 1.74x_2(-x_9 + (28.0x_{13} + x_2 + 1.24)(x_{14} - x_{16} - e^{\cos(x_1 - 2.01x_{11})} - 1.19) + 6.72) e^{x_2} + x_7(1.47 \cdot 10^4 - x_1) - x_7 - (-1.18 \cdot 10^3(x_1 - 3.88) e^{-x_9} - 4.30 \cdot 10^3) e^{x_{12}} + 434.e^{x_5} - e^{x_1 + x_{15}} + e^{x_1 + 2.06e^{x_{10}}} - e^{x_1 + x_{15} - x_6} + \log(x_4 + 0.0433) + 1.02 \cdot 10^4 \quad (32)$$

$$\hat{y} = x_{10} \cdot 1.35 \cdot 10^4 + (x_1 + x_{16})(x_{12} + x_7 + 1.74) \left(\frac{x_0}{-63.7} + (x_1 + x_8) (- (x_{13}x_9 \cdot 80.6 - (-x_{11} + x_{12}) 52.3) - 11.4) + 3.05 \cdot 10^3 \right) - (x_{15} + x_7) (-1.54 \cdot 10^3) - (x_5 - 3.39) (-651.) + ((x_{13} + x_6)(x_{11} \cdot 466. + e^{x_1} + 254.) - 398.) 5.02 - (x_{12} + x_{12} + x_6) 1.61 \cdot 10^3 \quad (33)$$

$$\hat{y} = -\frac{x_0}{x_{11} + x_{13} - \sin(x_1 + x_3(-0.587)) + \sin(x_1 + x_4 - 47.7) + 4.52} + x_1(x_1 + x_{11} + (x_{11} + x_8)(x_{13} + \sin(x_1)))(x_1 + x_{12} + x_6) \left(-x_9 - \frac{x_{11} - x_{14} \cdot 3.08}{x_{15} - 0.953} + 47.4 \right) + ((x_{10} + x_{12} + x_7) 1.01 \cdot 10^3 - 337.) 13.9 + \left(\frac{x_{12}}{x_7 - 0.384} + x_{16} \right) 2.88 \cdot 10^3 + \sin(x_1 + x_{13}) (-x_5 + x_8 - e^{x_{15}}) 969. + 1.64 \cdot 10^4 \quad (34)$$

$$\hat{y} = (x_1 + x_5 \cdot 0.216) \left(x_0(-0.0149) - (x_8 - x_9)(x_{13} + (x_{13} + x_{15} - 0.549) 3.29) (-70.7) + 2.04 \cdot 10^3 \right) (x_{10} + x_{10} + x_{12} + x_7 + e^{\sin(-(x_{16} + \sin((x_1 + x_{13} - x_8 + 1.59)(-0.346)(x_1(x_{15} - 0.167) - x_{11})) - 0.969) - 1.88)} + 1.27) \quad (35)$$

The text produced by the LLM in response to the first prompt E revealed several common parts:

1. $x_7 + x_{12} + x_{10}$ often appear together with a linear shift;
2. The same holds for $x_1 - x_{14} + x_{16}$;
3. Exponential interaction $e^{x_1 - x_9}$.

The answer to the second prompt produced the following generic equation:

$$\hat{y} = c_0 + c_1(x_1 - x_{14} + x_{16}) + c_2(x_{10} + x_{12} + x_7) - \frac{c_3 x_0}{c_4 + x_{12} + x_7} + c_5 e^{c_6 x_1 - x_9 + \frac{c_7}{\sin(x_2)}} + c_8 \log(x_0) \quad (40)$$

After a first attempt at constant optimization using CMA-ES [2], a few extra constants were added to multiply x_{12} and x_7 when they appear at the denominator:

$$\hat{y} = c_0 + c_1(x_1 - x_{14} + x_{16}) + c_2(x_{10} + x_{12} + x_7) - \frac{c_3 x_0}{c_4 + \mathbf{c}_9 x_{12} + \mathbf{c}_{10} x_7} + c_5 e^{c_6 x_1 - x_9 + \frac{c_7}{\sin(x_2)}} + c_8 \log(x_0) \quad (41)$$

Final equation, with optimized constants:

$$\begin{aligned} \hat{y} = & -\frac{89.49x_0}{-280.988x_{12} - 312.119x_7 + 1043.105} + 4133.879x_1 + 12806.192x_{10} + \\ & 12806.192x_{12} + 4133.879x_{14} + 4133.879x_{16} + 12806.192x_7 \\ & - 639.127e^{-964.96x_1 + \frac{2818.088x_9}{\sin(x_2)}} - 832.179 \log(x_0) + 6088.817 \\ R2 = & 0.8990 \end{aligned} \quad (42)$$

The value $R2 = 0.8990$ is lower than the original mean $R2 = 0.9449$, but it can still be considered as a high-quality solution; and on the other hand, the equation is considerably simpler than the equations in 30-39. A 10-fold cross-validation shows that optimizing the equation's constant on the whole dataset only leads to a modicum amount of overfitting, with mean $R2 = 0.8909 \pm 0.0296$.

E.4 Task 361264, default hyperparameters

Task 361264 (socmob) is a peculiar case, where no algorithm performs significantly better than the others, probably due to a large variance between folds; even using enhanced hyperparameters or hyperparameter tuning does not seem to significantly improve performance, so for this case we select the equations that PySR discovered with its default hyperparameters.

$$\hat{y} = x_4 x_4 (-0.00122) + x_4 \cdot 1.09 - (x_2 - 1.45) - \left(\frac{x_4}{7.94 - x_1} + \frac{x_4}{0.733 - x_1 x_3} \right) 0.171 \quad (43)$$

$$\hat{y} = -\frac{x_3}{-x_2 - 0.111} + 0.875 \left(x_4 + \frac{0.443}{(x_1 - 4.13) \frac{1}{\frac{x_4}{4.66 - (x_1 - 3.38)} - 0.537}} \right) \quad (44)$$

$$\hat{y} = x_4 + \frac{x_4 \left(0.711 - \frac{0.850}{x_1 + x_1 - 7.90} \right)}{(-1)1.74 \frac{1}{x_1 - 0.573} (x_1 - 8.05)} + 1.20 \quad (45)$$

$$\hat{y} = x_3 \left(2.76 - x_4 (-0.167) \frac{1}{x_1 - 7.93} \right) + \frac{x_4 \left(1.27 + \frac{0.310}{x_1 - 0.598} - \frac{0.486}{3.55 - x_1} \right)}{1.43} \quad (46)$$

$$\hat{y} = -(x_3(-(-9.98)x_2 - 10.4) - x_4 - 65.8) - \frac{5.36 \cdot 10^4}{821. - x_4} \quad (47)$$

$$\hat{y} = x_3 + \frac{x_3}{(x_1 - 0.562)(x_2 + 0.0690)} + x_4 \left(-1.00x_4 + x_4 - 1.03 + \frac{0.309}{x_1 - 7.87} \right) \quad (48)$$

$$\hat{y} = (5.99 - x_2)(x_3 + x_4(0.162 - x_4x_4 \cdot 2.47 \cdot 10^{-7})) \quad (49)$$

$$\hat{y} = x_4 \frac{1}{x_1 - 4.10} \cdot 271. \frac{1}{x_4 \cdot 0.185 + (x_1 - 8.02)(-798.)} - (x_2 - x_3)2.62 + (x_4 + 3.60)0.910 \quad (50)$$

$$\hat{y} = -1.46 \cdot 10^{-6}x_4x_4x_4 + x_4 + x_4 \cdot 0.216 \frac{1}{x_1 - 7.89} \quad (51)$$

$$\hat{y} = x_4 \left(1.16 - 0.00133 \left(x_4 - \frac{7.28}{8.01 - x_1} + \frac{47.1}{0.194 - x_1} \right) \right) \quad (52)$$

The text produced by the LLM in response to the prompt in E highlights the following points:

1. x_4 is the primary driver, often appearing in the form $c \cdot x_4$;
2. x_1 often appears as the denominator of a fraction in the form $\frac{1}{x_1 - c}$, with two recurring values $c_1 \approx 0.8$ and $c_2 \approx 0.6$
3. x_2 and x_3 often appear in linear terms.

After a deeper human analysis, we noticed that the equations for 3 folds were just functions of x_1, x_4 . Thus, we tried a first equation:

$$\hat{y} \approx c_1x_4 + \frac{c_2x_4}{x_1 - c_3} \quad (53)$$

Optimizing the constants c_1, c_2, c_3 using CMA-ES produced the following equation:

$$\hat{y} = 0.896x_4 - \frac{0.358x_4}{x_1 - 3.422}, \quad R^2 = 0.7536 \quad (54)$$

Assessing the overfitting with a 10-fold cross-validation, we obtained a mean $R^2 = 0.7299 \pm 0.1045$, which is close to the original value of $R^2 = 0.7661 \pm 0.0908$ found by Symbolic Regression over the more complex Eqs. 43-52. Adding another linear term in x_3 to Eq. 53:

$$\hat{y} \approx c_1x_4 + \frac{c_2x_4}{x_1 - c_3} + c_4x_3 \quad (55)$$

And optimizing again the constants with CMA-ES, produced the following equation

$$\hat{y} = 6.325x_3 + 0.936x_4 - \frac{0.24x_4}{x_1 - 3.701}, \quad R^2 = 0.7813 \quad (56)$$

Optimizing the constants in a 10-fold cross-validation gives a mean $R^2 = 0.7597 \pm 0.0859$. Adding a further linear term c_5x_2 produces no further improvements.

E.5 Task 361272, default hyperparameters

In task 361272 (`fifa`), Symbolic Regression’s performance is not significantly different from the other algorithms, already with default hyperparameters. The equations identified by PySR are:

$$\hat{y} = -x_3 + ((x_4 - 55.6)(x_4 + x_4 - 134.)(-0.463x_{15} + x_{15} + x_4 + x_4 + x_4 - 201.) + 2.79 \cdot 10^3) 1.13 \quad (57)$$

$$\hat{y} = (x_4 - 51.8)(x_8 + (x_4 - 64.4)(-x_0 + x_4(-x_6 - 38.9)(71.4 - x_4)0.000881 + x_5) + 171.) \quad (58)$$

$$\hat{y} = (x_4 - 53.6)x_5(x_4 - 53.6)0.0134(28.0 - (6.35 - x_9)(x_4 - 63.2)(x_4 - 64.5)0.00249) \quad (59)$$

$$\hat{y} = (x_8 + (x_4 - 55.5)(54.8 - x_4)(-3.18))((x_4 - 60.1)(x_4 - 66.8)x_9 \cdot 0.000894 + 9.77) \quad (60)$$

$$\hat{y} = (x_4(x_4 \cdot 7.06 - 912.) + 2.96 \cdot 10^4)(x_4 + x_6 \cdot 0.139 - 56.5) \quad (61)$$

$$\hat{y} = (x_4(x_4x_4 \cdot 0.0359 - 443.) + 1.91 \cdot 10^4) \left(-\frac{x_{27}}{x_6} + x_4 - 51.1 \right) \quad (62)$$

$$\hat{y} = \left(x_{15} + \frac{x_4 - 61.3}{0.145} \right) (x_4 - 64.9) \left(x_4 - 60.5 - \frac{-x_0 + \frac{x_{14}}{x_{22}} + 22.7}{\frac{1}{-17.1}x_{15}} \right) - 2.37 \cdot 10^3 \quad (63)$$

$$\hat{y} = x_4(-x_0(x_0 \cdot 0.593 - 29.6) + x_{15} + x_4(x_4 \cdot 7.89 - 1.43 \cdot 10^3) + 8.53 \cdot 10^4) - (x_{13} - 0.111) - 1.71 \cdot 10^6 \quad (64)$$

$$\hat{y} = -x_3 + (x_4 - 51.9)((-x_0 + x_2)3.93 + (x_4 - 64.5)(x_{15} - x_4(-7.30) - 531.)) \quad (65)$$

$$\hat{y} = (x_4 - 51.1)(x_4 - 62.9)(x_4 - 62.9)7.11 \quad (66)$$

The text produced by the LLM in response to the prompt claims the following similarities:

1. x_4 is the only variable that appears in every single equation across all 10 folds;
2. The most common mathematical pattern is the use of binomial offsets in the form of $(x_4 - C)$, with $C \approx 60$;
3. The common structural blueprint here relies on deeply nested multiplicative blocks;
4. Other variables have low frequency of appearance, with x_{15} (3/10 folds) being the most common.

In response to the second prompt, the LLM produces the following equation:

$$\hat{y} \approx -c_1x_3 + c_2(x_4 - c_3)^n \left[c_4x_4(c_5x_4 - c_6) + c_7x_k + c_8 \right] \quad (67)$$

with x_k which could be x_{15} , x_0 , or x_6 . The equation, however, contradicts some of the previous statements in the text. After a manual review, we settle on the tentative expression:

$$\hat{y} \approx c_0(x_4 - c_1)(x_4 - c_2)(x_4 - c_3)(x_4 - c_4) + c_5 \quad (68)$$

Optimizing the constants with CMA-ES provides the following equation:

$$\hat{y} = 0.17(x_4 - 56.201)^4 + 1823.898, \quad R^2 = 0.7568 \quad (69)$$

Which is a somehow surprising result, with constants $c_1 \approx c_2 \approx c_3 \approx c_4$. Optimizing the constants with a 10-fold cross-validation delivers a mean $R^2 = 0.7543 \pm 0.0312$, indicating only a small amount of overfitting.

E.6 Task 361257, default hyperparameters

In task 361257 (diamonds), no algorithm is significantly better or worse than others, mainly due to a large standard deviation in performance over the folds. Considering Symbolic Regression with default hyperparameters, the equations obtained are:

$$\hat{y} = (x_6 - 3.35) (x_0 (x_0 (-108.) + x_3 x_4 (x_2 (-1.00) + x_6) + 1.24 \cdot 10^3) - 23.8) \quad (70)$$

$$\hat{y} = -x_1 - x_6 (x_0 (x_0 (x_3 (x_2 (x_3 + 22.0) - 187.) + x_8) + x_1 (-24.7) - 621.) + 376.) + 1.13 \cdot 10^3 \quad (71)$$

$$\hat{y} = x_3 + (x_3 + x_6 (x_0 (x_1 - 55.0) - 51.6) + 151.) (x_3 (-x_2 + x_6 - 0.358) - x_6 + 21.8) + 473. \quad (72)$$

$$\hat{y} = ((x_7 - 2.16) (x_0 \cdot 65.6 - 35.8) - 54.7) (-0.291x_1 + x_1 - x_3 (x_2 - x_7) - x_6 - x_6 + 29.7) \quad (73)$$

$$\hat{y} = x_0 (x_3 + x_7) (x_2 (-47.8) - 961.) - 2.00 \cdot 10^3 \quad (74)$$

$$\hat{y} = x_0 ((-x_2 + x_7) 199.x_3 - 6.19 \cdot 10^3) - 2.41 \cdot 10^3 \quad (75)$$

$$\hat{y} = (x_0 - 0.278) (x_0 x_3 (x_2 (-201.) - 1.58 \cdot 10^3) - 4.94 \cdot 10^3) \quad (76)$$

$$\hat{y} = x_0 (x_6 - 3.82) (x_3 (514. + x_2 (-68.5)) + 1.18 \cdot 10^3) \quad (77)$$

$$\hat{y} = (x_0 - 0.315) (x_6 - 1.79) (x_3 x_4 (-x_2 + x_7) + 814.) + 652. \quad (78)$$

$$\hat{y} = (x_0 \cdot 287. - 88.9) (x_3 (-x_2 + x_7) + 19.2) \quad (79)$$

The text answering the prompt in E lists the following considerations:

1. While not universally present, x_6 and x_7 frequently appear, and they seem to perform a highly similar—or interchangeable—role within the models;
2. In several equations, x_0 acts as a direct, standalone multiplier for the entire core expression, taking the form $x_0 \times (\dots)$;
3. In others, the algorithm applies a simple affine shift to x_0 before multiplying it against the rest of the terms, creating patterns like $(x_0 - 0.278)$ or $(x_0 \cdot 287. - 88.9)$;
4. x_3 modulates an inverse relationship (or subtraction) between x_2 and either x_6 or x_7 ;

The text then proposes the following common skeleton:

$$\hat{y} \approx f(x_0) \cdot [x_3 \cdot (x_{6|7} - x_2) + \dots] \quad (80)$$

After a human intervention, the equation has the new structure:

$$\hat{y} \approx (x_0 - c_0) \cdot (c_1 x_3 \cdot (c_2 x_6 - c_3 x_2)) \quad (81)$$

Optimizing the constants using CMA-ES provides the following equation:

$$\hat{y} = 14.15 x_3 (x_0 - 0.305) (-22.734 x_2 + 38.959 x_6), \quad R^2 = 0.7191 \quad (82)$$

In a 10-fold cross-validation, we obtain a mean value of $R^2 = 0.7191 \pm 0.0068$, with no indication of overfitting.

However, this was one of the few cases where the default choice of PySR was giving a higher mean R^2 . What happens if we do the same with the other set of equations?

1. x_0, x_2 , and x_3 are present in all 10 equations. This indicates they have the highest feature importance for your target variable
2. x_6 and x_7 are secondary but highly prevalent, appearing in 9 out of the 10 folds
3. x_4 is rarely selected, appearing only in three folds
4. form of $x_3(-x_2 + \text{variable}) + \text{constant}$ or $x_3(C_1 + x_2 \cdot C_2) + C_3$
5. x_0 is a multiplier

$$\hat{y} \approx (x) = (c_1 x_0 + c_2) \cdot (x_j + c_3) \cdot \left[c_4 x_3 (c_5 x_2 + c_6 x_k + c_7) + c_8 \right] \quad (83)$$

After constant optimization with CMA-ES:

$$\hat{y} = (21.035x_0 - 3.686)(x_6 - 1.788)(0.605x_3(-4.107x_2 + 3.456x_6 + 4.696) + 40.006), \quad R^2 = 0.9618 \quad (84)$$

Optimizing the constants in a 10-fold cross-validation returns a mean $R^2 = 0.9618 \pm 0.0023$, indicating little to no overfitting.

E.7 Task 361254, default hyperparameters

In task 361254 (sarcos), SR performs worse than the tree-based ensembles, but its mean $R^2 = 0.9272$ is quite high and not that distant from the best, XGBoost with $R^2 = 0.9759$. In this case, we consider the equations discovered by PySR with default hyperparameters, as the gain with enhanced hyperparameters is relatively small:

$$y = -x_{10} + x_{14}x_2 + (38.6 + x_{20}(-0.320)) \left(x_0 + \frac{x_{17}}{56.9} \right) + \frac{x_{10}x_8 + x_{14} + x_{14} + 16.2}{0.785} \quad (85)$$

$$y = -x_{10} + (x_0 \cdot 14.6 + x_{14} + 7.63) 2.58 + \frac{x_0x_{16} - x_{10}x_7 + x_{17}}{1.60 - x_6} \quad (86)$$

$$y = x_0 \cdot 36.7 + x_{14} + \left(x_{18}(-0.164) - x_6 - -11.1 + \frac{-x_{10} + x_{17} + \frac{x_{14} + x_3 + 3.21}{0.462}}{x_3} \right) 1.15 \quad (87)$$

$$y = x_0 \cdot 38.0 - x_{18} \cdot 0.171 - \frac{x_7x_8}{0.352} + \frac{x_{14} + \frac{x_{14} + x_{14} + x_{17}}{x_3}}{0.864} - 0.0428 - -20.2 \quad (88)$$

$$y = -x_{10} + x_{12}x_{13} + x_{14} + \frac{x_{17}}{x_3} + (x_0 + 1.71)(x_{14} - (x_8 \cdot 2.02x_7 - 38.1)) - 40.1 - 4.27 \quad (89)$$

$$y = \frac{x_0}{0.0272} - x_{10} - x_{14}(-x_2 - 0.931) + x_{14} - x_8x_7 \cdot 2.89 - 2.00 + 21.8 + \frac{x_{14} + x_{17}}{x_3} \quad (90)$$

$$y = x_0 \cdot 36.1 - x_{10} - x_3(x_6 + 5.37) + (1.81 - x_3 \cdot 0.722)(x_{14} + x_{14} + x_{17}) - \frac{-x_{14} - 23.4}{0.810} \quad (91)$$

$$y = -(-x_0 + x_{10}) + \frac{-x_0x_3(-(-0.135)x_{20} - 8.72) + x_{14} + x_{17} \cdot 0.286 + x_4 + x_5 + 8.26}{0.396} \quad (92)$$

$$y = x_{14}(-35.2) + x_{17}(x_6 - -0.485) + (x_0 + x_{14})(x_2 + 37.8) - (x_7x_8 - x_7(0.601 - x_{10})) - -19.2 \quad (93)$$

$$y = -x_{10} - \left(-\frac{x_6}{-0.281} + \frac{-(x_{14} + x_{14} + x_{17}) - 11.4}{x_3} \right) + (x_1 + x_{14} + (28.4 - x_0)(x_0 + 0.385)) 1.25 \quad (94)$$

The text produced by the LLM answering the prompt in Textbox E lists the following common elements:

1. x_0, x_{14}, x_{17} appear in all equations;
2. The term $C \cdot x_0$ appears often, with $C \in [36, 39]$;
3. $-x_{10}$ appears in 9/10 equations;
4. There is a recurring nonlinear sub-structure where x_{14} and x_{17} are added together, often to form a numerator, and frequently divided by x_3 ;
5. When x_7 and x_8 appear, they generally do so as a multiplied pair (x_7x_8);

The text then presents the following generic equation:

$$\hat{y} \approx 37.5x_0 - x_{10} + \alpha x_{14} + \beta \left(\frac{2x_{14} + x_{17}}{x_3} \right) - \gamma(x_7x_8) + C \quad (95)$$

After a first human intervention, mainly focused on adding more constants to be optimized, the candidate equation becomes:

$$\hat{y} \approx c_0x_0 - x_{10} + c_1x_{14} + \left(\frac{c_2x_{14} + c_3x_{17}}{x_3} \right) - c_4x_7x_8 + c_5 \quad (96)$$

Optimizing the constants using CMA-ES on the whole dataset gives:

$$\hat{y} = 37.795x_0 - x_{10} + 1.149x_{14} - 2.975x_7x_8 + 20.163 + \frac{2.273x_{14} + 1.098x_{17}}{x_3}, \quad R2 = 0.9313 \quad (97)$$

What is remarkable in Eq. 97 is that the constant $c_0 = 37.795$, which multiplies x_0 , indeed has a value in the interval $[36, 39]$, as previously stated by the text. The $R2$ value is also quite high; optimizing the constant values in a cross-validation gives a mean $R2 = 0.9312 \pm 0.0026$, indicating a minimal amount of overfitting.

E.8 Task 361256, default hyperparameters

In task 361256 (cpu_activity), SR with default hyperparameters performs significantly worse than RF and XGBoost, but the difference in mean $R2$ is quite small, with $R2_{SR} = 0.9713$ and $R2_{XGB} = 0.9815$. We thus decided to analyze the equations obtained with default hyperparameters, as they tend to be smaller in size:

$$y = \frac{\frac{x_{17} - 1.71(x_{15} + x_{18})}{-39.0} - 75.1}{x_2 \cdot 17.5 \frac{1}{x_{20}} + 0.769} \quad (98)$$

$$y = x_{15}(-0.0627) - x_{17} \cdot 0.0227 + (x_2 + (x_0 + x_{16})7.55)(-0.00176) - 97.9 - \frac{9.08 \cdot 10^3}{x_{20} + x_5 + 90.3} \quad (99)$$

$$y = \left(-7.60 \cdot 10^{-6} + \frac{109.}{x_2(x_6 + 25.2) + x_{20} - (x_{15} + x_{17})(-477.)} \right) \left(x_{19}(-3.25) \frac{1}{-0.461x_{14} - 0.334} + x_{20} \right) \quad (100)$$

$$y = -(-0.0224) \left(x_{15}(-2.89) - x_{17} + x_2(-0.0827) + \frac{x_5}{-0.0348} \right) + 97.4 - \frac{1.65 \cdot 10^3}{\frac{1}{5.75}(x_{20} + x_5 - 95.0)} \quad (101)$$

$$y = \frac{x_0(-0.0229) - x_{16} \cdot 0.0432 + x_5(-2.29) + (x_{15} + x_9)(-0.135) + 171.}{x_2 \frac{1}{x_{20}} \cdot 46.9 + 1.75} \quad (102)$$

$$y = \frac{x_{20}}{-\frac{x_0}{-0.419} + \frac{x_2 - \frac{x_{20}}{x_5 - 37.4}}{2.50} + (x_{10} + x_{13} + x_{15} - x_5 - 77.7)8.80} + 0.947 \quad (103)$$

$$y = \frac{(0.532 + x_{17}(-0.000157))167. + \left(\frac{x_{16}}{0.532} + x_4\right)(-0.00387)}{0.948 - \left(-x_2 + \frac{86.0 - x_{15}}{0.0332}\right)\frac{1}{x_{20}} \cdot 23.2} \quad (104)$$

$$y = 0.0856 + \frac{53.2 - x_5}{0.547 + (x_{15} \cdot 41.9 + x_2 + (x_0 + x_{17})(x_{18} + 3.95))13.3\frac{1}{x_{20}}} \quad (105)$$

$$y = \frac{x_{17} + \frac{x_{15} + x_9}{0.427} - \left(-0.0643x_2 + \frac{x_4x_6}{-29.5}\right)}{-37.2} + 97.3 - \frac{9.39 \cdot 10^3}{x_{20} \cdot 1.10 + 94.0} \quad (106)$$

$$y = -0.233(x_{10} \cdot 0.233 + x_6 - 0.724) - \frac{2.22}{(-0.244)\left(0.100 - \left(x_{17} + \left(\frac{x_2}{-56.0} - -64.5\right)(-3.73)\right)(-42.8)\frac{1}{x_{20}}\right)} \quad (107)$$

The text produced by the LLM following the prompt in Textbox E claims to have found the following similarities:

- The variables x_2 and x_{20} are present in every single equation;
- The variable x_{15} appears in 9 out of 10 folds. The variable x_{17} appears in 8 out of 10 folds;
- The variable x_5 appears in 5 folds. The variable x_0 appears in 4 folds. The rest ($x_4, x_6, x_9, x_{10}, x_{13}, x_{14}, x_{16}, x_{18}, x_{19}$) appear sporadically, likely acting as fine-tuning parameters or fitting noise for their specific cross-validation folds;
- In the majority of the folds, x_{20} appears in a denominator, very frequently coupled with x_2 via multiplication;
- x_{15} and x_{17} consistently act as linear drivers.

The candidate equation summarizing the considerations is:

$$\hat{y} \approx (c_1x_{15} + c_2x_{17}) + \frac{\text{Numerator}}{c_3x_2\frac{1}{x_{20}} + c_4} \quad (108)$$

Interestingly, the Numerator part is not explicit, but we can gather from the context that it is likely meant to be either a constant or a function of the other sporadically appearing variables. After a first human intervention, the equation becomes:

$$\hat{y} \approx (c_0x_{15} + c_1x_{17}) + \frac{c_2}{c_3x_2 + \frac{1}{x_{20}} + c_4} \quad (109)$$

Optimizing the constants using CMA-ES gives the following equation:

$$\hat{y} = -0.058x_{15} - 0.03x_{17} + \frac{0.633}{1.354 \cdot 10^{-7}x_2 + \frac{1}{x_{20}} + 0.006}, \quad R^2 = 0.9718 \quad (110)$$

Several interesting details emerge from Eq. 110. First, the contribution of x_2 seems to be extremely small, with the constant set at a value $c_3 \approx 10^{-7}$. Then, the R^2 value is basically at the same level as the original. Evaluating the new equation in a 10-fold cross-validation gives a mean 0.9708 ± 0.0068 , indicating a small amount of overfitting. Interestingly, removing the term c_3x_2 from Eq. 109 and optimizing the constants again in a cross-validation gives a mean $R^2 = 0.9520 \pm 0.0105$, showing that the role of x_2 is not negligible.

E.9 Task 361617, default hyperparameters

In task 361617 (energy_efficiency), while SR has the worst result, with default hyperparameters its mean $R^2 = 0.9547$ is already extremely high, with XGBoost showing the best performance at $R^2 = 0.9988$. While SR's performance increases with enhanced hyperparameters, we decided to analyze the equations obtained with default hyperparameters, as the difference in performance is relatively small, and these equations tend to be smaller in size:

$$y = x_4 x_6 (7.91 + x_6 (-8.47)) + (x_4 - 2.71) (x_2 (x_3 \cdot 0.000217 - 0.00866) - x_3 \cdot 0.0579 + 5.02) + 5.58 \quad (111)$$

$$y = \frac{x_4 (x_2 \cdot 0.00217 + x_6 + 0.563) - \frac{0.0226}{0.217 - (x_0 - 0.586)}}{0.298} - 4.16 + \frac{0.0102}{x_6 - 0.00260} \quad (112)$$

$$y = \frac{x_0 (0.0150 x_2 x_6 - 4.31) + x_4}{0.160} + 4.15 - \frac{0.0883}{x_0 - 0.806} - \frac{0.812}{x_3 (0.0959 - x_6)} \quad (113)$$

$$y = (x_4 (x_6 + 0.263) + x_4) \left(x_2 \cdot 0.00681 - -1.09 - \frac{0.0665}{x_7 - -0.100} \right) - 1.84 - 1.13 \quad (114)$$

$$y = x_4 \left(x_2 \cdot 0.00378 + x_6 - \frac{0.166}{x_6 (-x_1 x_6 + x_3 + x_4) - -0.707} \right) 3.14 - 1.61 \quad (115)$$

$$y = x_2 (-0.0333) + (1.83 - x_0) 1.97 - (x_4 - -1.05) \left(-(x_2 \cdot 0.0132 + x_6 \cdot 2.73) - \frac{0.0730}{x_7 - 0.110} \right) \quad (116)$$

$$y = x_6 + \left(-x_0 + x_4 + \frac{5.41}{x_2 \cdot 0.440 - x_3} \right) (x_2 \cdot 0.00919 + x_6 \cdot 4.38 - -0.838) \quad (117)$$

$$y = -x_0 + (0.840 - x_4) (x_2 (-0.00721) - (x_6 \cdot 3.94 - -1.90)) - \frac{0.0838}{x_0 - 0.803} - \frac{0.607}{0.147 - x_7} \quad (118)$$

$$y = x_2 (x_4 - (2.01 - x_4 x_6)) 0.0113 - -5.63 - \frac{14.6}{-0.414 x_2 + x_3 - x_4} \quad (119)$$

$$y = -0.00800 (-x_4 - 1.06) \left(x_2 (0.140 x_4 + x_6) + \frac{3.14}{x_7 - 0.0373} \right) - -5.52 \quad (120)$$

The text generated by the LLM in response to the prompt in Textbox E claims the following common parts:

1. x_2, x_4, x_6 : These three variables are universally present in 100% of the folds;
2. x_0, x_3 : These appear in exactly 50% of the folds (5 out of 10);
3. x_7 : Present in 40% of the folds;
4. Across almost every single fold, x_2 appears with an incredibly small coefficient, usually on the order of 10^{-2} to 10^{-4} ;
5. There is a massive structural consensus that x_4 and x_6 interact multiplicatively, often acting together as the primary slope driver of the equation;
6. Whenever x_0 is used in a fraction, a mathematical pole (singularity) exists precisely around 0.80, with terms like $\frac{1}{x_0 - 0.80}$;
7. Whenever x_7 is selected by a fold, it is exclusively placed in the denominator of a localized rational term.

The equation proposed at the end of the text is:

$$\hat{y} \approx \beta_1 \cdot x_4(\epsilon x_2 + x_6) + \frac{\beta_2}{x_0 - 0.80} + \text{Correction}(x_3, x_7) + C \quad (121)$$

The text states that ϵ is a tiny positive scaling coefficient (≈ 0.001) and β_1 is a primary macro weight. Despite the claims of the last equation, a manual inspection of Eqs. 111-120 reveals that x_4 often multiplies the substructures function of x_0 and x_7 . After a human intervention, the new equation becomes:

$$\hat{y} \approx c_0 x_4 \left(c_1 x_2 + c_3 x_6 + \frac{c_4}{x_0 - c_5} + \frac{c_6}{x_7 - c_7} \right) \quad (122)$$

Optimizing the constants on the whole dataset with CMA-ES gives:

$$\hat{y} = -0.116 x_4 \left(-0.098 x_2 - 26.977 x_6 - \frac{0.307}{x_7 - 0.045} + \frac{0.082}{x_0 - 0.8} \right), \quad R2 = 0.9543 \quad (123)$$

Interestingly, the substructure $\frac{c_4}{x_0 - c_5}$ indeed features $c_5 = 0.8$. The $R2$ value is extremely close as the original, and a 10-fold cross-validation returns a mean $R2 = 0.9536 \pm 0.0078$, revealing only a small amount of overfitting.

E.10 Task 361236, enhanced hyperparameters

In task 361236 (auction_verification), SR performs significantly worse than the other algorithms, but still obtains a mean $R2 = 0.9588$, not far from the best result of $R2 = 0.9983$ by XGBoost. The equations obtained by SR during the cross-validation are:

$$y = x_4 + (x_2 + \cos(x_0)) \left(-x_0 \left(\frac{1.49 \cdot 10^3 x_2 e^{\cos(x_6)}}{x_4 (-1.80 e^{\cos(9.47 x_4)} + 2.17 e^{-x_5}) - 0.990} + 0.209 x_4 (x_4 - 1.74) e^{-x_2} e^{\left(\sin(1.78 x_4 - \log(x_4) - 1.55 + \dots \right)} \right) \right) \quad (124)$$

$$y = x_5 + \left(\left((-43.6 \sin(x_0 - x_2 + 2x_5 + 3.70) + 40.5 \cos(x_2 x_4 x_6^2)) \right) \left(4.26 x_3 - 4.26 \sin(x_4 + 0.697) + 4.26 \sin(22.0 x_4 + e^{\cos((x_2 - \dots \right)} \right) \right) \quad (125)$$

$$y = -x_2(x_2 + 0.544) - x_5 + \left(e^{(\sin(x_0) + \sin(x_0 + 0.226) - 2.23) \sin(-x_5 + \sin(x_5 - 0.734)) + e^{x_3}} + e^{x_2(\sin(x_0) + \sin(x_0 + 0.129)) + x_2 + e^{\sin(x_3(x_2 - \dots \right)} \right) \quad (126)$$

$$y = x_0 \left((x_6 + x_6) \left(x_2(-14.0) \frac{1}{-x_3 - x_6(-x_2 + x_3 - x_5) - 0.988} + 14.9 \right) + \left(-(-3.79) x_4 e^{x_3} + \left(103. - \left(x_5 - \frac{\dots}{\cos(-(x_3 - \dots \right)} \right) \right) \right) \quad (127)$$

$$y = (x_2 + \sin(x_5))(x_3 + 3.33)(x_2 \cos(x_5) + x_2 \cos(x_5(x_2 - 0.181) + 130.) + x_2 + 0.495) 1.36 \left(x_0 \left(x_0 x_0(-24.2) x_0 - x_2(\cos(x_2 - \dots \right) \right) \right) \quad (128)$$

$$y = -x_0 - (-x_3 - x_3 - 0.306) + e^{x_2+x_5-x_6-(0.302-x_3)} - \frac{\left(x_4 - \left(-x_0 - \frac{x_4(-123.) + \left((-1)1.09 - \frac{1}{-x} \right)}{(x_4x_5 - ((x_6+0.224) - \frac{123.}{0.506x_4})22.8)} e^{\frac{x_4(-123.) + \left((-1)1.09 - \frac{1}{-x} \right)}{(x_4x_5 - ((x_6+0.224) - \frac{123.}{0.506x_4})22.8)} \right)} \right)}{0.49} \quad (129)$$

$$y = \left(\frac{x_0 \left(2.20x_3 + e^{x_0+e^{0.630 \sin(x_5)}} - \sin(\cos(x_4)) \right)}{x_0 e^{x_0x_5} e^{-e^{x_3+\log(x_4) \sin(4.37x_5)}} - 0.694} + 1.46 \right) \left(-0.944x_5e^{x_6} + (x_2 - 0.290) \left(x_6 + e^{3.67 \sin(6.61x_4 - \frac{0.779x_5}{\cos(x_4)} - \sin(x_0))} \right) \right) \quad (130)$$

$$y = \frac{x_4 \left(x_5 - \left(\frac{x_3 + \sin \left(x_5 + e^{\frac{x_1}{x_1+x_5-\sin(x_6)+0.115}} + e^{e^{\cos(x_5 \cdot 1.08) - 0.0960}} \right)}{-0.365} + 4.74 \right) + 1.99 \right) e^{\frac{\sin \left(\frac{\sin(x_0)}{0.468} \right)}{0.433}} + e^{(\sin(x_5) + \cos(x_5) + 2.59)e^{\sin(x_0) + 0.0397}}}{0.459e^{\frac{1}{x_5 + \cos((x_4+x_4)(-14.2) - 3.41)} \left(x_2(-3.37) + \frac{x_2}{-x_2+x_4} + x_4 \right) \frac{x_2}{x_5((x_4-x_6x_6+x_6+2.90)0.619+0.768) + \sin \left(\frac{x_4}{1.89} \right) - 1.56}}} \quad (131)$$

$$y = \left(x_3 \sin(x_0) 0.329 e^{\sin((x_3+x_5)2.96)} + \sin(x_0 + 0.0397) \right) \left(x_1 + e^{x_5} (-35.8) - e^{x_2 \sin(-x_2+x_5+0.433)2.57 \frac{1}{0.613}} - \sin(x_6 \cdot 1.14) e^{-\frac{1}{x_6}} \right) \quad (132)$$

$$y = x_1 + \frac{\left(\frac{x_0 - 1.86}{x_4 + 0.579} + \sin(x_0) \right) \left((-x_0 + 7.46 + \frac{2.21}{x_5 - 3.77}) x_2 \cdot 15.4 \left(x_3 (9.15 - x_0) - \frac{3.26}{x_5 x_6 (x_2 - x_6) - 0.953} \right) + e^{x_2 - x_5 x_5} x_2 \cdot 82.1 \left(x_3 \sin(x_0) 0.329 e^{\sin((x_3+x_5)2.96)} + \sin(x_0 + 0.0397) \right) \right)}{0.49} \quad (133)$$

Answering the prompt in Textbox E, the text produced by the LLM claims the following structural similarities:

1. The variables x_0, x_2, x_3, x_4, x_5 , and x_6 appear in every single equation;
2. The variable x_1 is notably absent from several folds (specifically folds 0, 3, 4, and 6);
3. Every equation makes extensive use of the sin function;
4. The exponential function (e^x) is a universal structural component across the dataset;
5. The log function appears occasionally (e.g., in folds 0, 1, and 6), but unlike the trigonometric and exponential operators, it is not a universally required building block.

The LLM this time did not produce a generic form for the equation at the end of the first prompt, so the second prompt was used. The text produced by the LLM describes a complex structure:

$$\hat{y} \approx \mathcal{L}(\mathbf{x}) + \mathcal{M}(\mathbf{x}) \cdot \psi(\mathbf{x}) + c_1 e^{\mathcal{E}(\mathbf{x})} + \frac{\mathcal{N}(\mathbf{x})}{\mathcal{D}(\mathbf{x}) - c_2} \quad (134)$$

with:

$$\mathcal{L}(\mathbf{x}) = c_3x_0 + c_4x_2 + c_5x_3 + c_6x_4 + c_7x_5 + c_8 \quad (135)$$

$$\mathcal{M}(\mathbf{x}) = \sin(c_9x_0 + c_{10}) \quad \text{or} \quad (c_9x_2 + \cos(x_0)) \quad (136)$$

$$\psi(\mathbf{x}) = (x_2 + \cos(x_0)) \quad \text{or} \quad \sin(x_0) \quad \text{or} \quad (x_3 \sin(x_0)c_9 \cdots + \sin(x_0 + c_{10})) \quad (137)$$

$$\mathcal{E}(\mathbf{x}) = c_{11}x_2 \cos(x_5) + c_{12}x_3 + c_{13}x_6 \quad (138)$$

$$\mathcal{N}(\mathbf{x}) = c_{14}x_0 + c_{15}x_4 \quad (139)$$

$$\mathcal{D}(\mathbf{x}) = c_{16}x_3 + c_{17}x_5 + c_{18}x_6 \quad (140)$$

Setting $\mathcal{M}(\mathbf{x}) = \sin(c_9x_0 + c_{10})$ and $\psi(\mathbf{x}) = \sin(x_0)$ to then optimize the constants using CMA-ES gives the following result:

$$\hat{y} = 2934.956x_0 + 1838.41x_2 + 1900.943x_3 + 51.009x_4 - 997.463x_5 + \frac{7583.144x_0 + 11.245x_4}{-0.125x_3 - 2.662x_5 + 5094.609x_6 + 3.401} + 0.027e^{3.369x_2 \cos(x_0)} \quad (141)$$

While the expression looks obviously quite complex, optimizing its constants in a 10-fold cross-validation gives a mean $R2 = 0.7197 \pm 0.0385$, indicating a strikingly low amount of overfitting. Nevertheless, $R2 \approx 0.72$ is quite low when compared to the original $R2 \approx 0.96$, so either the LLM was not able to correctly detect the most meaningful similarities, or the equations are too different from each other to be reconducted to a common shape.

Looking back at Table ??, PySR with default hyperparameters obtained a mean $R2 = 0.8773$, so we deemed it useful to also examine the simpler equations obtained in that case:

$$y = (x_5(x_3 - 6.95) + 49.1) \left(\frac{x_0 \left(x_3 \cdot 106. + 112. + \frac{124.}{1.26 - x_5} \right)}{x_0 - 0.00899} + 16.4 \right) \quad (142)$$

$$y = (x_0(x_0(-6.31) - x_5 + 20.0) - 0.564) \left(-x_2x_4 + (x_3 - 3.92) \left(-x_5 + 168. + \frac{92.1}{1.27 - x_5} \right) \right) \quad (143)$$

$$y = x_0 \left(x_3 \cdot 37.5 + \frac{x_4 + \frac{\frac{x_3}{-0.0380} - 384.}{x_0 + x_5 + 0.153}}{x_5 - 1.25} \right) 55.3 - 1.17 \cdot 10^3 \quad (144)$$

$$y = x_0 \left(822. - \frac{535.}{x_5 - 1.27} \right) \left(x_3 - x_5 + \frac{11.6}{x_0 - 0.130} \right) \quad (145)$$

$$y = \frac{(x_0(4.28 \cdot 10^3 - (x_4 + x_5 \cdot 16.3)21.4) + 0.875) \left(\frac{x_3}{0.466} + 2.44 + \frac{2.35}{1.29 - x_5} \right)}{x_0 + 0.0105} \quad (146)$$

$$y = x_0(1.94 \cdot 10^3 - x_0 \cdot 642.) \left(x_3 \cdot 2.96 - x_5 - 6.40 - \frac{3.82}{x_5 - 1.27} \right) + x_4 \quad (147)$$

$$y = x_0 \left(x_3 \cdot 2.52 \cdot 10^3 + \frac{3.40 \cdot 10^4 - x_0(x_4 - 48.4)95.9}{x_2 + x_5(-1.09)x_5x_5} \right) - 1.11 \cdot 10^3 \quad (148)$$

$$y = (x_3 + 3.27) \left(x_0 \left(x_4(x_3 - x_5) + \frac{2.10 \cdot 10^3 - \frac{1.80 \cdot 10^3}{50.5 - (x_5 + 49.2)}}{x_0 + 0.147} \right) + 2.46x_4 \right) \quad (149)$$

$$y = -x_4 + (3.11 - x_0)(x_3 - x_5 + 7.70) \left(x_0 \left(173.x_3 + 401. + \frac{346.}{1.27 - x_5} \right) - x_6 - -34.6 \right) \quad (150)$$

$$y = \left(x_0 \left(x_0 + \frac{1.16 \cdot 10^3}{x_0 + 0.0315} \right) - x_4 + 122. \right) \left(x_3 \cdot 3.41 - x_5 - 22.1 + 28.8 - \frac{4.34}{x_5 - 1.27} \right) \quad (151)$$

Answering again the prompt in Textbox E with the new files, the text produced by the LLM gives these extra insights:

1. The most striking structural similarity across almost all folds is the identification of a rational term with a denominator highly dependent on x_5 approaching a value between 1.25 and 1.3. This suggests the underlying data has a sharp peak, asymptote, or non-linear transition near this threshold;
2. Similarly, multiple folds identify a secondary instability or necessary feature transformation when x_0 is close to zero. This manifests as another rational term dividing by $(x_0 + \epsilon)$, where ϵ is a small constant;
3. The overall topology of the equations tends to favor a product-of-sums, with x_0 frequently acting as a primary global scaling factor that sits entirely outside the heaviest parenthesis blocks. Folds 2, 3, 5, and 6 all explicitly factor x_0 to the very front of the main expression, multiplying the rest of the generated terms;
4. The evolutionary search frequently groups x_3 and x_5 together in linear combinations before passing them through further operations.

Again, a generic equation was not produced at the end of the first prompt, so the second prompt was used and the answer was:

$$\hat{y} \approx \left(c_1 x_0^2 + c_2 x_0 + \frac{c_3}{x_0 + c_4} + c_5 x_4 + c_6 \right) \cdot \left(c_7 x_3 - c_8 x_5 + \frac{c_9}{c_{10} - x_5} + c_{11} \right) + c_{12} x_4 + c_{13} \quad (152)$$

Optimizing the constants with CMA-ES generates the following equation:

$$\hat{y} = 92.431x_4 + \left(25.733x_3 - 8.269x_5 + 51.408 + \frac{36.008}{1.286 - x_5} \right) \cdot \quad (153)$$

$$\cdot \left(-72.759x_0^2 + 220.591x_0 - 1.82x_4 + 131.606 - \frac{414.259}{x_0 + 274.477} \right) - 6061.062, \quad R^2 = 0.8909 \quad (154)$$

In a 10-fold cross-validation, the mean performance is $R^2 = 0.8901 \pm 0.0223$, indicating almost a negligible amount of overfitting. It is also interesting to notice that this value is slightly higher than the original $R^2 = 0.8773$. Due to PySR's default settings, no functions heavily employed in the first set of equations, such as $\sin$ or the exponential, were used: But the good performance of these simpler equations led us to believe that it might be possible to obtain even better results by using PySR's default settings and just add more operators. However, this did not work, with a mean $R^2 = 0.8808$. Another run with `maxsize=60` and `maxiterations=1000`.

E.11 Task 361237, enhanced hyperparameters

In task 361237 (`concrete_compressive_strength`) SR obtains a remarkable mean performance $R^2 = 0.8612$. Even if this good result is significantly worse than the result obtained by the two other algorithms, with XGBoost being the best at $R^2 = 0.9317$, we deemed it interesting to analyze the equations obtained:

$$y = \frac{2.69x_4 \log(x_0)}{0.246 - 0.282x_3} + \frac{\log(x_5) - 0.775}{-29.0 \sin(x_7) - 13.9} - \frac{e^{\sin\left(\frac{\cos(\log(x_7(x_2 - x_3 x_7)(-x_4 - 0.877)) - 0.683)}{\sin(\log(x_0) - 0.306)}\right)} - \sin(1.97e^{\sin(1.36x_3 - \sin(x_1 x_3))})}{\cos(\sin(x_1 \sin(x_3)))} + 2.69 \log\left(\frac{x_0}{-1.41x_4 + \cos(x_2 + \sin(4.33x_4)) + \frac{x_3}{x_0}} - 1.36x_4 + \sin(x_3)\right) \quad (155)$$

$$y = 0.0707x_0 + \sin(x_3) - \sin(1.40x_3) - \sin(x_4) - \sin(0.268x_1 x_4) - 12.2 + \frac{0.292x_0 \sin\left(\frac{x_0}{\frac{1.36x_3}{-1.41x_4 + \cos(x_2 + \sin(4.33x_4)) + \frac{x_3}{x_0}} - 1.36x_4 + \sin(x_3)}\right)}{\cos(\sin(x_1 \sin(x_3)))} \quad (156)$$

$$y = \frac{2.54x_0 \left(e^{-\sin \left(\log \left(x_6 + x_7 \left(\frac{2.78x_0 (\log(x_7(x_2x_5x_6^2 + 0.00209))) - \cos(1.71x_3) + \cos(\sin(x_5x_6)) + 1.33e^{-\sin((x_7 + 0.0589)\log(x_6) + \sin(\sin(x_6)))})}{x_3} \right) + (3x_1 + x_2)e^{-\sin(x_5x_6)}} \right)} \right)}{x_3} + (3x_1 + x_2)e^{-\sin(x_5x_6)}} \quad (157)$$

$$y = x_4 + \sin(x_4 - x_7) + \sin(x_5 - \sin(-x_1 + x_3)) + \sin(-(-0.868)x_2 + x_3) + \cos(x_4) + \cos(x_7) + \cos\left(\left(x_1 + \frac{x_7}{0.543}\right)(-0.17x_2 + x_3)\right) \quad (158)$$

$$y = 0.0892x_0 - 0.158x_3 - \left(-\frac{x_1}{x_3} + \cos(\log(0.532x_0x_7)) - 4.53\right) \log((x_7 - 0.750)(0.0296x_0 + x_1 + 0.480x_2 - 0.480x_4)) + \log(x_3) \quad (159)$$

$$y = 0.105x_0 + 0.0187x_1 \log(x_7) + 0.00532x_2 - 0.194x_3 + x_4 - 0.00532(x_4 - \log(x_7))(-0.0690x_4(-x_3 + \cos(x_4)) - x_7) + 7.2x_5 \quad (160)$$

$$y = 0.0958x_0 + \left(\log \left(\frac{(x_1 + 11.6)(x_4 + 1.43)(x_1 \log(x_1 + 0.399) + x_5) \log(x_0x_2 + x_6 \left(0.515 + \frac{\sin((x_3 - x_6 + \log(x_0 + x_1) - 0.133)\sin(x_5x_6))}{-x_4 - 1.96} \right))}{x_3} \right) \right) \quad (161)$$

$$y = \left(-\cos(\log(3.92x_4 + 0.136)) + \frac{2x_0 + 2x_1 + x_2}{x_3} \right) \log \left(x_7 \left(x_3 + \frac{(x_4x_7 + x_6) \log(x_2 + 0.0346)}{x_3} \right) \right) + (0.0102x_1 + 0.0102x_5) \quad (162)$$

$$y = -0.0277x_7 + 1.50 \log \left(x_7^5 \left(x_0 + \frac{x_1^3 x_7^2}{x_3} \right) (x_2 + 2.76) \left(\frac{x_0}{x_3} + x_1 + x_2 \right) \right) - \sin((3.77x_1 - 3.77x_6) \cos(x_5)) - 1.50 \sin(x_1 - x_2) \quad (163)$$

$$y = x_0 \left(0.0293 + \frac{14.7}{x_3} \right) + \frac{-25.8 + \frac{2.22x_7 \left(-\frac{0.109x_0x_1e^{\sin(x_5 - \cos(\log(199.x_1 + 199.x_7)))} \sin(0.0547x_6)}{x_1 - 0.0741} + 0.766x_1 + \frac{0.766(0.477 - 0.576x_4)(0.453x_1 - 3.06)}{-0.478x_1 + 0.687x_2 - \sin(0.0901x_5)} \right)}{0.0741x_7 + 0.873}}{x_3} \quad (164)$$

After prompting the LLM with the content of Textbox E, the response remarks the following structural similarities:

1. The variables x_0, x_1, x_2, x_3, x_4 , and x_7 are selected and actively utilized in all ten of the provided equations. While x_5 and x_6 are extremely common, they are not strictly universal; for instance, x_5 is omitted entirely from Fold 1, and x_6 is excluded from Fold 5;
2. Every equation heavily relies on $\sin$ and $\cos$ operators to model non-linear periodic behaviors or introduce complex oscillations;

3. The genetic programming tree frequently nests these functions deeply, creating highly non-linear transformations;
4. The log function is used across almost all folds to compress features or handle exponential scaling;
5. The most striking structural similarity is how often the variable x_3 appears in the denominator;
6. The equations consistently feature standalone scalar biases added or subtracted at the root level of the equation;
7. The variable pairs (x_1, x_3) and (x_0, x_3) are frequently coupled together algebraically, interacting through addition, multiplication, or division before being passed into non-linear transformations.

No equation was produced at the end of the text, so the second prompt was used, obtaining:

$$\hat{y} \approx c_0 + \sum_{i \in \{0,1,2,4,7\}} c_i x_i + \frac{f_{\text{lin}}(x_0, x_1, x_2)}{c_{d1} x_3 + c_{d2}} + c_l \log(c_{l1} x_7 + c_{l2}) + \sum_k c_{s,k} \sin(\theta_k) + \sum_m c_{c,m} \cos(\phi_m) \quad (165)$$

Where the phase arguments θ_k and ϕ_m take either a linear or nested form:

$$\theta, \phi \in \{c_a x_j + c_b, \quad c_a x_j + c_b \sin(c_p x_q)\} \quad (166)$$

And $f_{\text{lin}}(x_0, x_1, x_2)$ is a linear function. A manual rewriting of this extremely complex expression gives:

$$\begin{aligned} \hat{y} \approx & c_0 + c_1 x_0 + c_2 x_1 + c_3 x_2 + c_4 x_4 + c_5 x_7 + \frac{c_6 x_0 + c_7 x_1 + c_8 x_2}{c_9 x_3 + c_{10}} + \\ & + c_{11} \log(c_{12} x_7 + c_{13}) + c_{14} \sin(c_{15} x_1 + c_{16} x_3) + c_{17} \cos(c_{18} x_1 + c_{19} x_3) \end{aligned} \quad (167)$$

After using CMA-ES to optimize the constants, we obtain:

$$\begin{aligned} \hat{y} = & 0.038x_0 + 0.01x_1 - 0.047x_2 - 0.017x_4 - 0.072x_7 + 13.513 \log(0.647x_7 + 1.877) + \\ & + 0.479 \sin(17.019x_1 + 11.752x_3) - 0.602 \cos(22.689x_1 + 44.699x_3) - 41.483 + \\ & + \frac{19.59x_0 + 20.475x_1 + 30.256x_2}{1.697x_3 - 32.863} \quad R2 = 0.8242 \end{aligned} \quad (168)$$

Optimizing the constants in a 10-fold cross-validation gives a mean $R2 = 0.8218 \pm 0.0286$, indicating a relatively low amount of overfitting; however, the standard deviation is also high, when compared to other tasks; and the performance of Eq. 168 is still lower than the original $R2 = 0.8612$, and not that much better than the mean result obtained by SR using default hyperparameters, $R2 = 0.8092$. It could thus be interesting to also analyze the simpler equations obtained with the original settings:

$$y = (x_0 - x_1(-0.761))0.111 + (-(-0.0411)x_3 - 11.8) \left(-0.0124x_2 + \frac{x_3}{x_7 - 17.9} - 6.95 \right) - 15.8 \quad (169)$$

$$y = \left(0.632 - \frac{0.365}{x_7} \right) \left(\frac{x_3}{(-1)0.150(x_7 + 20.6)} + x_4 + \frac{\frac{x_0}{0.637} + x_1}{11.0} + 33.1 \right) \quad (170)$$

$$y = \left(x_1 + \left(-x_0 + x_3 + (x_3 + x_4 + x_4) 42.1 \frac{1}{x_7 + 18.7} \right) (-1.47) + \frac{17.5}{x_4 - 0.123} \right) 0.0667 + 43.5 \quad (171)$$

$$y = -8.24 - 16.5 + \frac{x_0 \cdot 20.8 + x_2 x_5 \cdot 0.0162 + \left(x_1 - \frac{6.55 \cdot 10^3}{x_7 - 14.3} \right) 16.4}{x_3} - \frac{x_2}{x_0} \quad (172)$$

$$y = 19.2 - \frac{643.}{x_7 + 14.7} + \frac{x_0 \left((x_1 + x_2) \frac{1}{2.68} \cdot 0.0746 + 16.1 \right) + \left(\frac{x_1}{0.552} + x_2 \right) 3.87}{x_3} \quad (173)$$

$$y = x_0 \cdot 0.108 + x_1 \cdot 0.0811 + \frac{x_2}{17.7} + x_3(-0.241) - 57.9 - \frac{666.}{x_7 - -15.4} \quad (174)$$

$$y = \frac{x_0}{9.36} + \frac{x_1}{12.3} + \frac{x_2}{17.6} - \left(-(-0.145) x_3 + \frac{x_3 + 1.70}{\frac{1}{3.39}(x_7 + 14.5)} \right) + 39.9 \quad (175)$$

$$y = \frac{x_2}{1.37 + \frac{547.}{x_7}} + \frac{\frac{\frac{x_0}{-0.790} - x_1}{-17.9}}{\left(x_3 + \frac{x_5}{x_0 \cdot 2.11 \frac{1}{x_3} + x_7 - 3.23} \right)} \quad (176)$$

$$y = x_2(-0.0409) - 9.08 + \left(x_0 + \frac{x_1}{1.25} + x_2 + 7.13 \right) 33.9 \frac{1}{x_3 - -24.8 + \frac{2.60 \cdot 10^3}{x_7 - -7.79}} \quad (177)$$

$$y = 15.9 + \frac{10.9}{\frac{x_7}{(-0.0823)(x_2 + x_2 + x_6)} - 0.254} + \frac{x_0 \cdot 18.6 + \frac{x_1 + x_2}{0.0739}}{x_3 - -0.932} \quad (178)$$

Answering the prompt in Textbox E with Eqs. 169-178, the LLM produces text highlighting the following similarities:

1. The variables x_0 , x_1 , x_3 , and x_7 are present in all 10 equations. This strongly suggests they are the primary drivers of your target variable;
2. The variable x_2 appears in 8 out of the 10 folds, missing only in folds 1 and 2;
3. Variables x_4 , x_5 , and x_6 appear very sparsely. x_4 only shows up in folds 1 and 2; x_5 in folds 3 and 7; and x_6 exclusively in fold 9. These are likely capturing fold-specific variance or noise;
4. Almost every equation places x_7 in a denominator, establishing an inverse relationship. Even more interestingly, it frequently takes the form of $\frac{\text{constant}}{x_7 + k}$ or a similar variation. The offset constant k is remarkably stable across the folds, typically clustering between 14 and 20;
5. Linear Combinations of x_0 , x_1 , and x_2 : These three variables generally act as the linear foundation of the model. They are often scaled by small constants and added together before being subjected to the more complex non-linear terms;
6. While x_7 is usually isolated in its own rational term, x_3 frequently acts as a global dividing factor for larger groupings of variables.

Again, no generic equation was returned, so the second prompt was given in input to the LLM, obtaining the following:

$$\hat{y} \approx c_0 + \frac{c_1 x_0 + c_2 x_1 + c_3 x_2}{c_4 x_3 + c_5} + \frac{c_6}{x_7 + c_7} \quad (179)$$

Optimizing the constants using CMA-ES generates the following expression:

$$\hat{y} = 13.342 - \frac{739.372}{x_7 + 17.038} + \frac{627.204x_0 + 477.672x_1 + 379.5x_2}{27.052x_3 + 801.264} \quad R2 = 0.8179 \quad (180)$$

And in a 10-fold cross-validation we obtain a mean $R2 = 0.8154 \pm 0.0272$, which is surprisingly close to the previous $R2 = 0.8218 \pm 0.0286$ of the much more complex Eq. 168! The two equations share the building block $\frac{c_1 x_0 + c_2 x_1 + c_3 x_2}{c_4 x_3 + c_5} + c_0$. In fact, this block alone, with constants re-optimized by CMA-ES, can obtain a mean $R2 = 0.4404 \pm 0.0455$.

E.12 Task 361258, enhanced hyperparameters

In task 361258 (kin8nm), SR manages to obtain a mean performance of $R2 = 0.7440 \pm 0.0435$, which is not separable from the values of RF and XGBoost: No algorithm performs significantly better or worse than the others. The equations found by SR during the 10-fold cross-validation are:

$$y = \sin \left((x_2 + 1.23) 0.197 + ((x_0 + (x_1 (-0.394) + x_4 \cdot 0.545 - x_4 (-1.13) - x_7 + x_7 - (x_1 + x_3 x_4) + (x_3 + x_4 + (x_1 + x_6) (-0.197) - 1.13) - 0.197) - 0.197) \right) \quad (181)$$

$$y = \cos \left(\sin \left(\left(\frac{x_6 (-x_3 - 0.802)}{-1.81} - (x_2 (-1.23) + (x_3 (-(-1.04) (x_5 - x_7) - 0.375) - x_4 (-0.477) - 0.262) \right) (-0.228) + \left(x_5 \right) \right) \right) \quad (182)$$

$$y = \cos \left(\sin \left(e^{\left(x_4 - (x_2 + x_2) - \left(\frac{\cos (x_0 + x_5 - ((0.911 - (x_2 + x_3)) (- (x_2 + \sin (x_5 + \frac{x_6}{0.651} - (-0.553 (x_3 + x_4 - (x_0 - (x_5 + x_5 + x_6 - \frac{x_7}{0.478})) - 0.795) + \cos (x_2)) + 0.786)) - 1.13) - 0.197) - 0.197) \right) \right) \right) \quad (183)$$

$$y = 0.646 - \left(-(-0.435) \left(-\left(\frac{x_4}{-0.308} - \left(x_6 \left(x_4 \left(-0.111 \left(0.999 - \left(\left(x_0 - \frac{x_2}{-0.341} - \left(x_2 - \left(\frac{x_2}{-0.507} - (e^{x_0} + 2.91) \right) \right) - \sin (x_5) \right) \right) \right) \right) \right) \right) \right) \quad (184)$$

$$y = \sin \left(\left(-10.5x_2 + \frac{x_7}{e^{x_7}} - \left(x_0 - \left(\frac{x_4 x_5 (-x_6 - x_6 + x_7) - (-x_2 + (x_5 + x_6) ((-x_1 - 0.508) (-0.471) \frac{1}{e^{x_7}} - (x_5 + x_6) (-1.13) - 0.197) - 0.197) \right) \right) \right) \quad (185)$$

$$y = \cos \left(\cos \left(-0.0385 \left(x_0 + \left(-0.383x_4 + \left(x_3 + ((-1.60 (0.252 - x_4) - 1.53) 1.47 + 2.16) \left(-(x_1 - (x_2 - 1.05) + \sin (1.71 - 0.197) - 0.197) \right) \right) \right) \right) \right) \quad (186)$$

$$y = \cos \left(\frac{\cos \left(\left(\left(\left(x_0 + x_2 - (x_3 + (\cos ((x_2 + x_5 + x_6 - (x_2 + x_2 + x_4)) (-0.426)) (-(-x_7 + 0.982 + 1.86) - 1.19) (-5.20) + \cos (x_5) - 0.197) - 0.197) \right) \right) \right) \right) \right) \quad (187)$$

$$y = \cos \left(\cos \left(x_2 \cdot 0.387 + \left(\left(x_0 - x_4 + \left(x_6 \left(x_4 (x_2 - 1.66) \left(x_6 + \frac{x_4 + x_5 - (x_4 - x_5 + x_7)}{0.671} \right) - (0.796 - (x_5 - 0.705)) \right) \right) \right) \right) \right) \quad (188)$$

$$y = (x_2 (-0.177) + (((0.623 - (x_2 + x_5) 0.396) 0.320 + \cos (x_1 (-1.01) + (x_0 + 0.230) x_2 (-0.240) - 0.478)) x_0 (-0.223) + (-x_5 - 0.197) - 0.197) \quad (189)$$

$$y = -0.0429x_0 + \left(\sin (x_2) + \sin \left(\left(x_4 (0.799 - x_2) - x_5 - x_6 + \frac{-(-x_5 - x_6 + x_7 + (1.40 - x_3) (1.40 - x_3) \sin (-1.56x_2 - 0.197) - 0.197)}{0.671} \right) \right) \right) \quad (190)$$

Answering the prompt in Textbox E, the LLM produces text highlighting the following structural similarities:

1. Variables $x_2, x_3, x_4, x_5, x_6, x_7$ appear in equations for all folds;
2. x_0 appears in 9/10 folds;
3. In almost every fold where it is isolated, the global coefficient for x_2 converges to approximately -0.155 to -0.164 ;
4. Similarly, x_0 consistently acts as a linear offset with a scaling factor hovering tightly around -0.044 ;
5. Variables x_5 and x_6 are nearly inseparable. They repeatedly appear summed together as an entry point into a trigonometric function;
6. Whenever the $x_5 + x_6$ block appears, x_3 and x_7 are typically nearby acting as phase shifts or multipliers inside the nested sin/cos expressions.

The text ends with the following generic equation:

$$\hat{y} \approx C_0 - 0.16x_2 - 0.044x_0 + g(x_3, x_4, x_5 + x_6, x_7) \quad (191)$$

where $g(\cdot)$ is a periodic function (or nested composition of cosines/sines) heavily dependent on the invariant sum $x_5 + x_6$, modulated by x_4, x_3 , and x_7 . After a human intervention, the equation now has the following shape:

$$\hat{y} \approx c_0 + c_1x_0 + c_2x_2 + \sin(x_6x_7(c_3x_3 + c_4x_4)) \quad (192)$$

But let's also take a look at the simpler equations generated by SR using default hyperparameters:

$$y = (x_4 - x_6 (x_3 + x_5 (x_4 + 0.504) + 0.591) - (x_2 + x_2 + (x_0 + x_3x_5) 0.647)) 0.0744 + 0.719 \quad (193)$$

$$y = 0.394 + \left(x_0 + x_5 + x_6 - (2.13 - x_2) \left(x_4 - x_6 \left(\frac{x_3}{1.59} + x_5 (x_4 - -0.415) \right) + 3.98 \right) \right) (-0.0383) \quad (194)$$

$$y = 0.732 - (x_2 + x_2 + (x_3 + 0.592) (x_5 + x_6 - -0.343) - (x_4 - (x_0 + x_5x_6) 0.554)) 0.0709 \quad (195)$$

$$y = 0.719 - \left(x_2 - -0.402 \left(x_3x_5 - x_4 + x_6 \left(x_3 - \left(\frac{x_5 \left(x_4 + \frac{0.458}{x_6} \right) - 0.727}{-0.708} \right) \right) \right) \right) 0.155 \quad (196)$$

$$y = (x_0 + (x_2 - x_4) 3.26 + (x_5 + x_6) (x_3 + (x_4 + 0.378) (x_5 + x_6 - x_7) + 0.898)) (-0.0466) - -0.747 \quad (197)$$

$$y = -0.152x_2 + ((x_3 + 0.743) (x_5 + x_6 - x_7 \cdot 0.534 - -0.416) - (-0.685x_0 + x_4 \cdot 1.21)) (-0.0564) + 0.733 \quad (198)$$

$$y = 0.737 - (x_2 \cdot 0.154 + (x_0 - x_4 \cdot 2.18 + (x_3 + x_5 (x_4 + 0.473) + 0.691) (x_5 + x_6 - x_7)) 0.0513) \quad (199)$$

$$y = 0.718 + (x_0 - x_2 (-3.04) + x_3 (x_5 + x_6) + x_4 (x_5x_6 - (1.38 - x_2)) + x_5 + x_6) (-0.0508) \quad (200)$$

$$y = (x_2 (-2.33) - (-x_4 - x_4 + (x_0 + x_1) 0.487 + (x_5 + x_6) (x_3 + x_4x_5 + 0.719))) 0.0636 + 0.730 \quad (201)$$

$$y = 0.727 - \left(-x_2 + x_4 + \left(x_3 + \frac{x_4 (x_5 + x_6 - x_7)}{1.47} - 0.637 \right) (x_5 + x_6 + 0.231) (-0.437) \right) (-0.150) \quad (202)$$

If we follow the equation that the LLM gives us:

$$\hat{y} \approx c_0 + c_1(c_2x_2 + c_3x_0 + c_4x_4 + (x_5 + x_6 + c_5x_7 + c_6)(x_3 + c_7x_4x_5 + c_8x_4 + c_9)) \quad (203)$$

Optimizing its constants with CMA-ES leads to the following equation:

$$\begin{aligned} \hat{y} = & -0.042x_0 - 0.153x_2 + 0.126x_4 - \\ & - 0.061(x_3 + 1.129x_4x_5 - 0.054x_4 + 0.667)(x_5 + x_6 - 0.611x_7 + 0.246) + 0.72, \\ R^2 = & 0.5648 \end{aligned} \quad (204)$$

The performance in a 10-fold cross-validation is $R^2 = 0.5640 \pm 0.0234$, higher than the mean $R^2 = 0.5241$ of the original SR with default hyperparameters. Taking an equation from the most complex set and optimizing the constants gives $R^2 = 0.7704$ with a mean $R^2 = 0.7699 \pm 0.0119$ in a 10-fold cross-validation.

$$\begin{aligned} \hat{y} = & -0.05x_0(-0.08x_2 - 0.08x_5 + \cos(1.142x_1 + 0.457x_2(x_0 + 0.779) + 0.566) + 0.466) - \\ & - 0.158x_2 + 0.02x_4 + 0.021(-1.541x_1 + x_4 + 0.393x_7 + 0.138(x_2 + x_7) \cdot \\ & \cdot (-2.315x_4(0.018x_1 + x_6) + 4.536) + 0.007(-54.579x_4 - 12.551)(-3.126x_1 + 12.264x_2 - 15.621) - \\ & - 1.541 \cos(x_3) + \cos(x_4) + \cos(24.105x_3(0.422 - 0.91x_4)) + 55.075) \cos(1.049x_5 + 1.02x_6 + 0.018x_7) - \\ & - 1.213 \sin(0.015x_0 + 1.051x_5 + 1.02x_6 + 1.533) - \\ & - 0.272 \cos(x_2 - 0.069) \cos(x_3 + x_5 + 0.954x_6 - 0.637x_7 + 3.552) + 0.689 \end{aligned} \quad (205)$$

So, in this case we can conclude that the LLM is failing to capture the common parts of the equations which actually matter; or the equations are so different from each other that the task is more difficult than usual.

Asking the LLM again after running the SR equations through a more refined simplification process based on equality saturation graphs from the `SymbolicUtils.jl`, gives the following form:

$$\hat{y} \approx c_0 + c_1x_0 + c_2x_1 + c_3x_2 + c_4x_4 + c_5 \cdot \sin\left(c_6 + c_7x_1 + c_8x_2 + c_9x_3 + c_{10}x_5 + c_{11}x_6 - c_{12}x_7\right) + c_{13} \cdot \cos\left(c_{14} + c_{15}x_3 + c_{16}x_5 + c_{17}x_6 - c_{18}x_7\right) \quad (206)$$

Optimizing the constants gives a mean $R^2 = 0.6187 \pm 0.0190$, in a cross-validation:

$$\hat{y} = -0.043x_0 - 0.025x_1 - 0.149x_2 + 0.071x_4 - 0.197 \sin(0.06x_1 + 0.165x_2 + 0.94x_3 + 0.928x_5 + 0.921x_6 - 0.608x_7 - 0.911) - 0.03 \cos(0.015x_0 + 1.051x_5 + 1.02x_6 + 1.533) + 0.689 \quad (207)$$

A manual merge of Eqs. 204 and 207 gives:

$$\begin{aligned} \hat{y} \approx & c_0 + c_1x_0 + c_2x_1 + c_3x_2 + c_4x_4 + \\ & + c_5 \sin(c_6 + c_7x_1 + c_8x_2 + c_9x_3 + c_{10}x_5 + c_{11}x_6 + c_{12}x_7) + \\ & + c_{13} \cos(c_{14} + c_{15}x_3 + c_{16}x_5 + c_{17}x_6 + c_{18}x_7) + \\ & + c_{24}(c_{25}x_5 + c_{26}x_6 + c_{19}x_7 + c_{20})(c_{27}x_3 + c_{21}x_4x_5 + c_{22}x_4 + c_{23}) \end{aligned} \quad (208)$$

Optimizing the constants with CMA-ES:

$$\begin{aligned} \hat{y} = & -0.043x_0 - 0.026x_1 - 0.154x_2 + 0.108x_4 + \\ & + 0.035(0.443x_3 + 1.556x_4x_5 + 0.07x_4 + 0.343)(-0.811x_5 - 1.535x_6 + 0.788x_7 + 0.046) - \\ & - 0.008 \sin(0.376x_1 - 0.088x_2 + 2.919x_3 + 3.735x_5 + 1.662x_6 + 4.165x_7 - 3.351) + \\ & + 0.161 \cos(1.004x_3 + 1.031x_5 + 0.969x_6 - 0.657x_7 + 0.69) + 0.691 \\ R^2 = & 0.6944 \end{aligned} \quad (209)$$

With a performance in a 10-fold cross-validation of $R^2 = 0.6939 \pm 0.0163$, indicating a small amount of overfitting.

E.13 Task 361259, enhanced hyperparameters

In task 361259 (pumadyn32nh), SR performs significantly better than the other two algorithms, showing a mean performance of $R^2 = 0.6585 \pm 0.0159$, with the second best being RF with $R^2 = 0.6474 \pm 0.0176$. The equations discovered during the 10-fold cross-validation are:

$$y = \cos(x_4)x_{15}(-0.00107) \quad (210)$$

$$y = x_{15} \cos(x_4)(-0.00107) \quad (211)$$

$$y = \left(x_{15} - \left(x_2 + x_4 + \frac{0.650}{x_{16}} \right) 4.87 (0.250 - x_{11} \cdot 0.105x_{28}) \right) \cos(x_4)(-0.00107) \quad (212)$$

$$y = \cos(x_4)(-0.000850) \frac{1}{-0.000169 - \frac{0.786}{x_{15}}} \quad (213)$$

$$y = \frac{\left(x_{15}(-0.000195) - \frac{6.57 \cdot 10^{-5}}{0.489 - \frac{1}{-((-x_{19} + x_4) + x_{19}(-0.430) \frac{1}{x_{16}})) \left(x_2 - e^{\sin(x_{15}(x_{10} + 0.206 - \frac{0.161}{x_9}))} - 0.790 \right) - 2.40} \right) \cos(x_4)}{0.182} \quad (214)$$

$$y = \cos(x_4)x_{15}(-0.00107) \quad (215)$$

$$y = ((-x_{28} + \cos(x_{16}))x_{20} \cos(x_5)x_{15} \cdot 0.0115 + x_{15} \cdot 0.779 \cos(x_4) - \sin(x_2))(-0.00141) \quad (216)$$

$$y = \left(-(-0.0103)x_{12}x_{30}(-x_{21} + x_{25}x_{29}) \left(x_5 + \frac{x_{31}}{x_{24}} \right) + x_{15} - x_4 \right) \cos(x_4)(-0.00108) \quad (217)$$

$$y = (x_{15} \cos(x_4) - \sin(x_2))(-0.00108) \quad (218)$$

$$y = -0.00108(x_{15} - x_4 - 0.222) \cos(x_4) \quad (219)$$

With a few exceptions, the equations are surprisingly simple, and even before starting the LLM-assisted analysis, it is possible to discern a few interesting patterns, such as the frequent occurrence of just x_4 and x_{15} and the use of transcendental functions, especially $\cos$. Nevertheless, using the approach illustrated in Section E, we prompt the LLM and obtain a text highlighting the following similarities:

1. Folds 0, 1, and 5 independently discovered the exact same, simplistic equation (just slightly rearranged in Fold 1): $\hat{y} = -0.00107 \cdot x_{15} \cdot \cos(x_4)$;
2. The more complex equations are essentially this baseline model with additional terms acting as additive or multiplicative adjustments. Instead of just multiplying x_{15} by $\cos(x_4)$, the longer equations substitute x_{15} with a more elaborate parenthetical group like $(x_{15} - \text{modifier})$ before multiplying by the core $\cos(x_4)$ and the constant factor;
3. x_4 and x_{15} appear in all equations. While not universal, x_2 appears in three of the more complex folds as a modifying term. Specifically, Folds 6 and 8 subtract $\sin(x_2)$ from the core equation;
4. There is a highly stable scaling factor distributed across the models.

The equation produced at the end of the text is:

$$\hat{y} \approx c_1 \cos(x_4) (x_{15} + c_2 \sin(x_2) + c_3 x_4 + c_4) \quad (220)$$

Which, however, seems too complex given the remarks above. Optimizing its constants using CMA-ES gives:

$$\hat{y} = -1.07 \cdot 10^{-3} (x_{15} - 1.116x_4 - 1.24 \sin(x_2) + 0.08) \cos(x_4), \quad R2 = 0.6622 \quad (221)$$

And in a 10-fold cross-validation, we obtain a mean performance of $R2 = 0.6620 \pm 0.0161$, indicating a relatively small amount of overfitting. Seeing that the form $\hat{y} \approx c_0 x_{15} \cos(x_4)$ is recurrent in multiple folds, optimizing its only constant c_0 gives:

$$\hat{y} = -1.077 \cdot 10^{-3} x_{15} \cos(x_4), \quad R2 = 0.6615 \quad (222)$$

With a mean performance of $R2 = 0.6612 \pm 0.0157$ over the 10-fold cross-validation, indicating again a minimal amount of overfitting.

E.14 Task 361267, default hyperparameters

In task 361267 (brazilian_houses), no algorithm is significantly better or worse than the others, due to a large standard deviation in performance between folds. SR still shows the highest mean $R2 = 0.6718 \pm 0.3171$ (with the second best, RF, at $R2 = 0.3828 \pm 0.3372$). Even when using enhanced hyperparameters, SR's mean performance only grows slightly to $R2 = 0.6937 \pm 0.2696$, so for this analysis we will focus on the equations found while employing default hyperparameters:

$$y = x_3 \cdot 1.29 \cdot 10^3 + x_4 \cdot 388. + x_8 + (x_0 + x_6) \left(x_1 + \frac{x_1 (x_0 - x_6)}{x_3} \right) + \frac{298.}{0.472 - x_7} \quad (223)$$

$$y = x_0 \cdot 342. + x_2 \left(x_3 (-6.23 \cdot 10^4) \frac{1}{x_1 - -38.7} - 916. + x_7 (-4.50 \cdot 10^4) \frac{1}{x_1} \right) + x_3 \cdot 1.79 \cdot 10^3 + x_8 \quad (224)$$

$$y = x_1 + x_2 \cdot 282. - x_4 (-498.) - x_7 \cdot 1.21 \cdot 10^3 - (x_0 (-390.) + (-(-1.21 \cdot 10^3) x_3 + x_8) (-1.00)) - 274. \quad (225)$$

$$y = x_3 \cdot 975. + x_8 - \left(x_5 + \frac{x_7}{0.00345} \right) + \left(\frac{x_1}{x_8 + 1.68} - x_8 (-871.) \frac{1}{0.996 (x_8 + 6.69 \cdot 10^3)} \right) 19.3 \quad (226)$$

$$y = x_3 \cdot 1.38 \cdot 10^3 + x_8 + (x_4 - x_7 - (-x_0 + x_7)) 448. \quad (227)$$

$$y = x_0 \cdot 328. + x_3 \cdot 1.28 \cdot 10^3 + x_8 + (-x_4 + x_7) (-654.) \quad (228)$$

$$y = x_0 x_4 \cdot 157. - x_3 (-1.56 \cdot 10^3) + x_8 \quad (229)$$

$$y = x_3 \cdot 963. + x_8 - \left(x_0 + x_2 - x_2 \cdot 304. \frac{1}{x_1 + 22.6} + x_4 - x_7 - x_7 + 4.96 \right) (-448.) \quad (230)$$

$$y = x_4 \left(1.04 \cdot 10^3 + \frac{x_3 (x_8 - 4.13 \cdot 10^4)}{x_1} \right) + x_8 - (-x_3 + x_7) 1.51 \cdot 10^3 - 1.53 \cdot 10^3 \quad (231)$$

$$y = -(-1.48 \cdot 10^3) x_3 + x_7 (-1.20 \cdot 10^3) + x_8 - (x_0 + x_4) (-430.) \quad (232)$$

The text produced by the LLM in response to the prompt in Textbox E highlights the following similarities:

1. The most striking similarity is that every single equation relies on the exact same linear foundation. If you strip away the more complex fractional and interaction terms, the core model is consistently:

$$y \approx C \cdot x_3 + x_8 + \text{residuals}$$

;

2. Frequency of appearance of the other variables: x_7 (9/10 folds), x_0 and x_4 (8/10), x_1 (6/10);
3. The models favor computing a heavy linear combination first, and then adding a smaller, non-linear component to capture variance. You rarely see deep nesting; the equations generally flatten out into sums of distinct terms;
4. When a variable recurs in a similar structural position, its coefficient tends to stay in the same order of magnitude. For example, when x_0 appears as a standalone linear term, its coefficient sits tightly in the 300s (342 in fold-1, -390 in fold-2, 328 in fold-5);
5. When division is used, it almost always takes the form of a constant divided by an expression (e.g., $\frac{298}{0.472-x_7}$ in fold-0) or a variable divided by another shifted variable (e.g., $\frac{1}{x_1-38.7}$ in fold-1). These act as localized asymptotes to correct specific data regions.

The generic equation summarizing the results is reported as:

$$\hat{y} \approx c_1 x_3 + x_8 + c_2 x_0 + c_3 x_4 - c_4 x_7 + \mathcal{F}_{\text{NL}}(\mathbf{x}) + c_0 \quad (233)$$

With $\mathcal{F}_{\text{NL}}(\mathbf{x})$ representing the heavily fold-specific interactions and rational components the symbolic regression used to fine-tune the curve. While the exact formulation changes per equation, $\mathcal{F}_{\text{NL}}(\mathbf{x})$ usually takes one of two generic forms: (i) asymptotic corrections, typically formulated as $\frac{c_k}{x_1-k}$ or $\frac{c_k}{k-x_7}$, with variable x_1 almost exclusively used in denominators to create local spikes or shifts in the data; (ii) simpler interactions, occasional direct multiplications of the secondary variables, such as $c_k(x_0 \cdot x_4)$.

In a first step, we decided to ignore the more complex interactions represented by $\mathcal{F}_{\text{NL}}(\mathbf{x})$ and just use CMA-ES to optimize the constants of:

$$\hat{y} \approx c_1 x_3 + x_8 + c_2 x_0 + c_3 x_4 + c_4 x_7 + c_0 \quad (234)$$

The result is quite surprising, because the equation with constants optimized on the whole dataset has an incredibly high R^2 :

$$\hat{y} = 356.76x_0 + 1389.623x_3 + 580.394x_4 - 1126.077x_7 + x_8 + 81.76, \quad R^2 = 0.9413 \quad (235)$$

However, in a 10-fold cross-validation the performance becomes lower at $R^2 = 0.7081 \pm 0.2373$, showing the same high variance observed for all ML algorithms. As a completely linear expression already shows signs of considerable overfitting, adding more terms is likely to be counterproductive. Instead, we perform a short ablation study, removing the terms one by one to assess their relative importance. The $c_4 x_7$ term has relatively little impact on the quality of the equation, which can then become:

$$\hat{y} = 393.222x_0 + 1399.873x_3 + 571.234x_4 + x_8 - 884.748, \quad R^2 = 0.9405 \quad (236)$$

With a performance in a 10-fold cross-validation corresponding to $R^2 = 0.7004 \pm 0.2379$.

E.15 Task 361260, enhanced hyperparameters

In task 361260 (miami_housing), SR performs significantly worse than the tree-based ensembles, showing a mean performance of $R^2 = 0.8152 \pm 0.0364$, with the best being XGBoost at $R^2 = 0.9223 \pm 0.0128$. Still, the R^2 value is indicating a model with a good performance. The equations discovered by SR during the 10-fold cross-validation are:

$$y = 4.42x_0 \left(-\frac{63.7x_{11}x_{14}x_3}{x_8} + x_3 \right) - x_{11} + x_{14} \left(-\frac{x_7x_8 - (x_{10} + x_4 + x_9)(x_{14}x_3 \log(x_{10}) + x_3(x_0 + 35.6) - x_9)}{\left(5.39 + \frac{2.88 \cdot 10^3}{x_{14}}\right) \left(\frac{15.0x_{13}x_9}{x_3} + x_3\right) \cos(x_{14})} + \frac{6.37 \cdot 10^3}{\dots} \right) \quad (237)$$

$$y = x_0 + x_2 - x_3 + x_5 - x_9 - (x_1x_0 \cdot 10.6 - x_{13}) + \frac{(\cos(e^{x_{14}} \cdot 10.6) + 2.72) \left(x_{14}(x_5 + 0.267) + (1.59 - x_1x_3)2.62 - (-x_0 + \dots) \right)}{\dots} \quad (238)$$

$$y = 3.46x_{10} + x_{14} (x_3 (2x_0 - 6.75) + x_4 - x_6 + 0.598) + 5.04x_2 + x_3 - 2.09x_4 + \frac{17.4x_5^2 (x_3 + 12.5)}{x_6x_8} + 2x_5 - 2.95x_7 - 2.09x_8 - \quad (239)$$

$$y = -x_{10} + x_3 \cdot 1.87 + x_5 - x_9 + (x_8 \cdot 31.8 - 0.220) \sin \left(\frac{30.1}{x_{11} \left(2.74 + \frac{\left(\frac{x_8}{x_{11} - \frac{x_7 - 7.07}{-6.84 \cdot 10^4}} + (x_2 + 0.0783 - 0.444)(-1.35) \right) \cos \left(-\frac{0.813 + \frac{1}{(1.42)}}{x_6} \right)} \right)} \right) \quad (240)$$

$$y = \frac{x_{14} \left(x_3 + x_3 + \frac{x_7 + (x_0 + 11.4)(x_2 + x_2 + x_3 + x_3) + 0.680 + \frac{x_3 \cdot 681 - (x_5 - 121 \cdot (x_{10} - 5.84 \cdot 10^3))}{\frac{1}{x_9} (x_{11} x_7 - x_5 + x_8)}}{-x_3 + x_3 + \frac{x_3}{x_0 + 13.2}} \right)}{\frac{1}{6.18(x_3 + x_3 + x_3 + x_3 + x_3)}} (x_{11} + x_3 + x_7 + x_9) - x_2 + x_3 \cdot 89.1 + x_4 + x_4 + x_7 + \frac{x_3 + x_3}{\cos(x_1)} \quad (241)$$

$$y = x_3 (1.08 - x_{11}) - x_9 + \frac{(x_0 + \cos(x_{12})) (x_8 - (x_0 x_1 (-x_1 + x_{10} + x_{10} + x_2 + x_4) - x_1 (x_0 - 1.99) (x_7 + (x_0 + x_2) (x_{12} - \dots)))}{-x_1 + x_7 - (-x_8 + (x_{10} + x_{11} + x_{14} x_8 + x_5 - x_6 - x_7 - x_8 - 0.960) 0.171)} \quad (242)$$

$$y = x_1 - x_{11} - x_{11} + x_{14} (x_3 - x_6) + x_{14} + x_3 + x_3 + \frac{x_4}{0.530} + x_5 + x_9 - \left(x_{14} - \left(\frac{x_1 x_{11}}{0.357} + 0.0251 \right) \right) - \frac{-(-159.) x_0 x_0 + x_3 (e^{\dots})}{(-1) 0.782 \cdot 1.3} \quad (243)$$

$$y = x_{14} (-x_3 - 98.2) (-11.1) (x_{14} x_3 + x_4 + x_5 - x_7) \frac{1}{x_7 + x_8} + \frac{x_{14} x_{14} (x_0 x_{14} + x_3)}{0.146} + x_2 - x_3 (-98.3) + \frac{x_0 (x_0 + x_0) + x_2}{\cos(x_1)} - \quad (244)$$

$$y = x_{11} (-3.87) + x_2 + x_2 + x_2 + \left(x_1 + \left((x_1 - x_{11} - x_{14} (x_0 - x_{10} + x_{11} - x_2 - x_4 + x_9 - (x_1 + x_5) - 6.05 \cdot 10^4) + x_3 \cdot 235. \right) \right) \quad (245)$$

$$y = -x_{10} - x_{11} + 2x_2 + x_5 + \frac{0.545 \left(-x_{11} \left(x_3 - \frac{4.55 \cdot 10^5 \log(x_2)}{x_6} \right) + x_{14} (x_{10} + 33.8x_3 + 33.8e^{x_{14}} - 4.14 \cdot 10^4) + x_4 + x_7 - 2x_9 \right)}{\cos(x_1 - \dots)} \quad (246)$$

Answering the prompt in Textbox E, the LLM returns text highlighting the following similarities:

1. The variables $x_0, x_1, x_2, x_3, x_4, x_5, x_8, x_9, x_{10}, x_{11}$, and x_{14} appear in all 10 folds. This indicates they hold the highest predictive power for your target objective and are essential to the model’s fitness;
2. The variables x_6 and x_7 are present in 9 out of 10 equations (missing only in Fold 4 and Fold 8, respectively);
3. The algorithm actively prunes x_{12} and x_{13} . x_{12} appears in 6 folds, while x_{13} is largely ignored, appearing only in Folds 0, 1, and 3;
4. Most equations feature a shallow, un-nested linear combination of variables added to the more complex fractions. For example, simple terms like $+x_2, +x_5, -x_9$, or $-x_{10}$ frequently appear at the outermost level of the expression tree (prominently in Folds 1, 2, 3, 4, 6, and 9). This suggests the target has a baseline linear relationship with these specific variables;
5. There is a recurring structural motif where x_1 governs the frequency or scaling of a periodic function, which is then used as a denominator to modulate other terms;
6. The variable x_{14} is repeatedly subjected to exponential or periodic transformations, serving as a complex scaling factor. We see $e^{x_{14}}$ explicitly in Folds 1, 3, and 9, and $\cos(x_{14})$ or $\sin(\dots x_{14} \dots)$ in Folds 0 and 8;
7. All folds exhibit deeply nested fractional structures. Variables like x_8, x_9 , and x_{14} repeatedly appear in the denominators of these deep sub-trees. For instance, Folds 0, 2, 5, and 7 feature blocks that heavily divide by x_8 or combinations involving it. This suggests the optimization is aggressively rewarding interaction ratios or inverse relationships for these specific variables.

No generic equation is produced at the end of the text; entering the second prompt gives the following equation and accompanying text:

$$f(\mathbf{x}) = (c_1x_2 + c_2x_3 + c_3x_4 + c_4x_5 - c_5x_9 - c_6x_{10}) + (c_7x_{14}(x_3 + c_8) + c_9e^{c_{10}x_{14}}) + \frac{\mathcal{N}(x_0, x_3, x_7)}{c_{11}x_8 \cos(c_{12}x_1 + c_{13}) + c_{14}} \quad (247)$$

With the numerator $\mathcal{N}(x_0, x_3, x_7)$ highly variable across folds, acting as a “catch-all” for remaining non-linear polynomial interactions. It typically involves combinations of x_0, x_3 , and x_7 , often interacting with logarithmic functions or nested sines depending on the specific fold.

After a human rewriting and reorganization of Eq. 247, we obtain:

$$\hat{y} \approx c_0 + c_1x_2 + c_2x_3 + c_3x_4 + c_4x_5 + c_5x_9 + c_6x_{10} + c_7x_{14}(x_3 + c_8) + c_9e^{c_{10}x_{14}} + \frac{c_{15}x_0 + c_{16}x_3 + c_{17}x_7}{c_{11}x_8 \cos(c_{12}x_1 + c_{13}) + c_{14}} \quad (248)$$

Optimizing the constants of Eq. 248 with CMA-ES gives:

$$\begin{aligned} \hat{y} = & 3.428x_{10} + 19.25x_{14} (x_3 - 276.342) + 3.698x_2 + 132.981x_3 + 2.174x_4 + \\ & + 3.332x_5 - 3.676x_9 + 0.002e^{3.707x_{14}} + 8225.43 + \\ & + \frac{-6674.663x_0 - 464.115x_3 - 3349.321x_7}{358.913x_8 \cos(1785.459x_1 - 1291.356) - 1192.136} \\ R^2 = & 0.6772 \end{aligned} \quad (249)$$

In a 10-fold cross-validation, the performance is $R^2 = 0.6746 \pm 0.0304$, indicating only a small amount of overfitting.

E.16 Task 361242, default hyperparameters

In task 361242 (superconductivity), SR performs significantly worse than the tree-based ensembles, with a mean $R^2 = 0.7113 \pm 0.0203$, with RF showing the best performance with $R^2 = 0.9217 \pm 0.0074$. Moreover, no improvements in performance are observed even when SR employs enhanced hyperparameters. For this reason, we decided to analyze the equations from the experiment with default settings. The equations discovered during the 10-fold cross-validation are:

$$y = \frac{x_{56}x_{70}}{x_{73} \left(\frac{x_{44}x_{44}}{(x_{29}-8.35)(x_{27}+x_{28}x_{35}(-0.580)-0.510)} + x_{74} - 1.23 \right)} + 7.01 \quad (250)$$

$$y = \frac{x_{66}}{0.0859} + x_9x_{70} \cdot 0.0112 \frac{1}{x_1 + x_{43}x_{78}} \left(\frac{x_{27}}{x_{72}} + x_{77} - 12.4 \right) \quad (251)$$

$$y = x_{25} + (-x_{44}x_{45} + x_{70} - (x_{27}(-0.485) - x_9 + 58.1))x_{26} \frac{1}{(-43.9)(x_{74} - 0.767)} (-15.2) + 6.47 \quad (252)$$

$$y = x_{16}x_{77} - x_{45} + \frac{-x_0 + \frac{x_{27} \left(\frac{x_{16}x_{26}x_{61}x_9}{49.5} + x_{70} \right)}{1.12 \cdot 10^3}}{x_{72} - 1.23} + 5.67 \quad (253)$$

$$y = x_{25}(x_{77} + 3.65) + \frac{\left(-0.103 + \frac{x_{70}}{x_{44}x_{73}} \right) \left(-x_{30} + x_{44} \cdot 35.3 \frac{1}{x_{22}} \right)}{-\frac{0.654x_{28}}{x_{44}} - 0.454} \quad (254)$$

$$y = -x_{35}x_{35} + \frac{\frac{x_{27}x_{70} \left(\frac{x_6x_9}{\frac{1}{0.0162}x_{73}} + 0.332 \right)}{x_{74}} - x_{65}(-1.89 \cdot 10^3)}{x_{23} - x_{44}(-0.928)} \quad (255)$$

$$y = x_{15} + \frac{x_{29}(x_6(-x_7 - 20.4)(x_{70} - 54.6)(-6.76 \cdot 10^{-5}) + x_{66})}{x_{71} - 0.203} \quad (256)$$

$$y = -(-x_6 - 3.74) + \frac{\frac{x_{56} \left(\frac{x_{44}}{-2.08} + \frac{x_{70}+x_9}{x_{72}} \right)}{(-0.715)x_{73} - 1.32}}{0.498} + x_{66} + x_{77} \quad (257)$$

$$y = -x_{25} + \frac{x_{26}(x_{70} + x_9 + x_9)}{x_{72}x_{73}} - \left(-\frac{103.}{x_{46} + x_{54}} + \frac{81.0}{x_{54}} \right) - 3.01 - \frac{317.}{x_1} \quad (258)$$

$$y = x_{30}x_{70} \cdot 85.6 \frac{1}{x_{28} - -53.1} \frac{1}{x_{72}(x_{44} - -12.1)} + x_{65}(x_{77} + 1.04)1.90 - x_{65} \cdot 4.50 \frac{1}{x_{64}} \quad (259)$$

The text that the LLM produces answering the prompt in Textbox E claims the following similarities:

1. x_{70} (The Core Driver): This is the most critical variable, appearing in all 10 equations. x_{44} : Present in 6 equations (folds 0, 2, 4, 5, 7, and 9). x_9 : Present in 6 equations (folds 1, 2, 3, 5, 7, and 8). x_{72} : Present in 5 equations (folds 1, 3, 7, 8, and 9). x_{73} : Present in 5 equations (folds 0, 4, 5, 7, and 8). x_{77} : Present in 5 equations (folds 1, 3, 4, 7, and 9). x_{27} : Present in 5 equations (folds 0, 1, 2, 3, and 5).

2. Almost every equation is built around a complex fractional term (a rational function) where a numerator is divided by a heavily nested denominator.
3. The models frequently add a constant or a simple standalone variable outside of the main fraction to shift the baseline. For example, the +7.01 at the end of fold-0, or starting with $x_{25} + \dots$ in fold-2 and fold-4.
4. When x_{70} appears, it almost exclusively dictates the behavior of the numerator. It is frequently multiplied by other variables to scale the primary rational term (e.g., $x_{56}x_{70}$ in fold-0, or x_9x_{70} in fold-1).
5. The denominators are highly non-linear, often featuring the subtraction of constants from specific variables (e.g., $x_{74} - 1.23$ in fold-0 or $x_{72} - 1.23$ in fold-3) to create asymptotic behavior.

The text ends with the following generic equation and the accompanying explanation:

$$\hat{y} \approx y = c_0 + c_1x_a + \frac{c_2x_{70}x_b + c_3}{c_4x_cx_d + c_5x_e + c_6} \quad (260)$$

Where c_0 represents the global constant offset. c_1x_a represents the standalone linear variable offset (commonly acting like x_{25} or x_{15} in your folds). The numerator is dominated by x_{70} multiplied by a scaling variable x_b (acting like x_9 or x_{56}) and a coefficient c_2 . The denominator captures the non-linear interaction of variables like x_{44} , x_{72} , or x_{73} (represented by generic variables x_c, x_d, x_e), scaled and shifted by constants c_4, c_5 , and c_6 .

After a human intervention, Eq. 260 is rewritten as:

$$\hat{y} \approx c_0 + c_1x_{15} + c_2x_{25} + \frac{c_3x_{70}x_9 + c_4x_{70}x_{56} + c_5}{c_6c_{44}x_{72} + c_7x_{73} + c_8} \quad (261)$$

Optimizing the constants of Eq. 261 with CMA-ES results in:

$$\hat{y} = 78.469x_{15} - 78.356x_{25} + 3.467 + \frac{60.046x_{56}x_{70} + 1.456x_{70}x_9 - 1794.169}{179.632x_{72} + 81.658x_{73} - 201.841} \quad R^2 = 0.6638 \quad (262)$$

In a 10-fold cross-validation, the mean performance is $R^2 = 0.6633 \pm 0.0153$. While this performance is satisfying, it is lower than the original SR's mean $R^2 = 0.7113 \pm 0.0203$. In an attempt to obtain an equation which includes more terms, we prompt the LLM again:

Additional LLM prompt used for this case study

Can you reformulate the equation including all terms that appear most often with the actual variables, for example: instead of $c_1x_a, c_1x_{25} + c_2x_{25}$ and so on? It's ok to have a more complex equation, I just want to capture as many recurrent terms as possible.

The text in response includes the following equation:

$$\hat{y} \approx c_0 + c_1x_{25} + c_2x_{77} + c_3x_{25}x_{77} + c_4x_{66} + \frac{x_{70}(c_5x_9 + c_6x_{27} + c_7) + c_8x_9 + c_9x_{27}}{(c_{10}x_{72} + c_{11})(c_{12}x_{73} + c_{13}) + c_{14}x_{44} + c_{15}} \quad (263)$$

Optimizing the constants of Eq. 263 with CMA-ES gives:

$$\begin{aligned} \hat{y} = & 4.94x_{25}x_{77} - 21.151x_{25} + 18.978x_{66} - 3.351x_{77} + \\ & + \frac{1010.188x_{27} + x_{70}(19.412x_{27} + 92.837x_9 - 2207.771) - 4663.478x_9}{54.219x_{44} + (123.637x_{72} - 216.296)(59.175x_{73} - 42.878) + 5527.946} + 11.917 \\ & R^2 = 0.7223 \end{aligned} \quad (264)$$

In a 10-fold cross-validation, the mean performance is $R^2 = 0.7217 \pm 0.0143$, indicating only a small amount of overfitting.

E.17 361247, default hyperparameters

On task 361247, SR performs significantly worse than the other two approaches. The performance with enhanced hyperparameters is actually even worse than with default hyperparameters, highlighting possible issues. The equations discovered by SR for the different folds are:

$$y = x_{11} \cdot 10.1 - 8.81 - \frac{0.00999}{7.50 \cdot 10^3 \frac{1}{x_8} \frac{1}{\frac{x_{11}x_4}{x_8(x_9 - 0.743)0.220} + x_8}} \quad (265)$$

$$y = x_8(-0.00297) + \left(x_{10} \cdot 2.89 \cdot 10^{-5} - \frac{0.00479}{x_{10}}\right) \left(x_7 \frac{1}{-0.00300x_{10}(x_5 - 0.113) + x_{10}}(-0.00472) + x_8\right) + 2.94 \quad (266)$$

$$y = \frac{\left(x_{11} - \left((x_0 + x_9 + x_8)x_9 \frac{1}{x_{12} + x_2}x_0 \cdot 1.52 + \frac{1.25}{x_{11}}\right)\right)(x_{13} - -4.14) + 2.01}{1.13} \quad (267)$$

$$y = 0.356 - \frac{(x_9 + 0.224) \left(\left(x_{11} - \frac{0.142}{x_9}\right)x_{11}x_{11}(-0.928) + x_8 \cdot 0.000671 + 0.276\right)}{0.417x_9} \quad (268)$$

$$y = -\frac{-(x_8(-0.000189) - (-x_{11} - 3.08) - 3.87) + \frac{0.788}{x_2}}{0.114} - 0.0331 + \frac{x_8 \frac{1}{x_9}(-0.000389) + 0.0807}{x_9} \quad (269)$$

$$y = -0.437 + \left(\frac{x_{10}}{0.375} + \frac{x_2}{1.07x_7} + 233.\right) 3.09 \frac{1}{x_8} \quad (270)$$

$$y = x_{11} \left(\frac{x_{11}(x_8 + (x_0 + 44.8)21.1)}{x_8} - 0.290\right) - 1.46 + \frac{2.12}{(-1.91)x_{10}} + \frac{x_{13}}{x_1} \quad (271)$$

$$y = x_{10} \cdot 0.520 \left(0.112 - x_3 \cdot 8.33 \cdot 10^{-6} \frac{1}{x_9}x_9\right) + (x_8 \cdot 0.000363 - 0.308)(-6.65) + \left(\frac{x_0}{0.134} - x_3\right) 4.79 \cdot 10^{-5} \quad (272)$$

$$y = \frac{x_{11}}{0.431} - x_8(-0.000636) \frac{1}{x_{11}(0.713 - x_{11})} + \frac{0.101}{\frac{1}{-2.95}x_9} \quad (273)$$

$$y = \frac{-(-0.0463)x_{10} - x_8 \left(x_4(-5.79 \cdot 10^{-8}) \left(\frac{x_3}{-1.80} - 0.0463\right) \frac{1}{x_3(x_9 - 0.587)} \frac{1}{x_9} - -0.00408\right) + 3.72}{1.62} \quad (274)$$

The text produced by the LLM highlights the following similarities:

1.

$$\hat{y} \approx c_0 + c_1x_{11} + c_2x_{10} + \frac{c_3x_{11} + c_4}{x_8(c_5x_9 + c_6) + c_7} \quad (275)$$

However, optimizing the constants of Eq. 275 as it is gives unsatisfying results with low R^2 . After expert intervention, the new equation looks like:

$$\hat{y} \approx c_0 + c_1x_8 + c_2x_{10} + c_3x_{11} + \frac{c_3x_{11} + c_4}{x_8(c_5x_9 + c_6) + c_7} \quad (276)$$

Optimizing the constants of Eq. 276 gives:

$$\hat{y} = 0.028x_{10} + 1.21x_{11} - 0.003x_8 + \frac{3304.762x_{11} - 3446.023}{x_8(3.953x_9 - 9.853) + 3678.592} + 1.133, \quad R^2 = 0.8022 \quad (277)$$

And in a cross-validation we have a mean $R^2 = 0.8020 \pm 0.0035$.

E.18 Task 361251, enhanced hyperparameters

On this task, SR is significantly worse than the other two approaches, but it gains considerably when passing from default to enhanced hyperparameters. The equations identified by SR with enhanced hyperparameters are:

$y = \sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$\sin$

$0.0279 \log$

$(x_{10} \log (x_2 + 0.675) + 0.208)$

$1.73x_{11}x_3$

(278)

[illegible]

[illegible]

$$y = 0.483 \sin \left(\sin \left(0.0167x_8 (x_8 + \log(x_3 - x_7)) \log(x_0) - 0.0681 \left(-0.896x_{11} + 0.00582(-x_2 + 2.15x_3x_8 + x_3 - 1.12) \log \left(\frac{x}{2} \right) \right) \right) \right) \quad (281)$$

[illegible]

$$y = -\sin \left(0.169 \sin \left(\sin \left(\sin \left(\sin \left(\sin \left(\sin \left(\sin \left(-0.0175x_0x_{10}x_2x_8 + (0.0330x_{11} (0.000117x_0x_2x_8 + 0.0410x_1 + 0.0 \right) \right) \right) \right) \right) \right) \right) \right) \right) \quad (283)$$

[illegible]

$$y = \sin \left(\sin \left(\sin \left(1.03 \sin \left(0.0231 \log \left(0.0226 x_1 x_3 (x_8 (x_0 - 0.619) + x_9) \right) \left(x_{11} - 0.0274 x_3 + 0.0244 \log \left(-\frac{50.6 e^{x_2 (x_{10} - x_{11})}}{(285)} \right) \right) \right) \right) \right)$$

$$y = \sin \left(\sin \left(0.0192x_9 + \frac{0.00448 \left(x_2x_8 + \cos \left(x_{11} + x_2 + \frac{0.891}{x_2} \right) \right)}{\cos(1.07x_{10})} - 0.0106 \log \left(\frac{0.539e^{-\left(x_9 + e^{\sin(x_{11}) - \frac{2.21}{x_3}} \right) \log(x_1x_{11})} e^{\frac{1.56}{x_3} + \frac{2.1}{x}}}{x_{10} \log \left(x_9 + e^{e^{-\frac{2.21}{x_3} + 0.539}} \right)} \right) \right) \right) \quad (286)$$

[illegible]

The text generated by the LLM claims the following common structural elements:

1. The variables $x_0, x_1, x_2, x_3, x_8, x_9, x_{10}$, and x_{11} are present in every single equation. This indicates that these 8 features hold the primary predictive power for your target variable y
2. The most glaring visual similarity is the heavy use of repeated, nested sine functions, i.e., $\sin(\sin(\sin(\dots)))$. This is a frequent artifact in symbolic regression when the algorithm tries to bound an output between $[-1, 1]$ or map a highly periodic or complex non-linear boundary layer
3. Every single equation constructs its inner logic using the exact same non-linear mathematical operators: natural logarithms ($\log$), natural exponentials (e^f), sines ($\sin$), and cosines ($\cos$)
4. Across nearly all expressions, x_0 repeatedly appears in the denominators of major fractional blocks. It acts as a universal scaling or normalization factor for the rest of the interacting inputs
5. You will notice specific feature pairings that constantly cling to each other across folds—for instance, x_{11} and x_2 are frequently combined within exponentials or logs, and x_1 often scales alongside x_9 .

The equation found by the LLM is:

$$\hat{y} \approx c_0 \sin \left(c_1 \sin \left(\dots \sin \left(\frac{c_2 \cdot e^{c_3 x_{11}} \cdot \log(c_4 x_2 + c_5)}{c_6 x_0 + c_7} + c_8 x_1 x_9 + c_9 x_{10} x_3 + c_{10} \right) \dots \right) \right) \quad (288)$$

With a nested set of sin functions of arbitrary length. After expert intervention, we settle on the following structure:

$$\hat{y} \approx c_0 \sin \left(c_1 \sin \left(\frac{c_2 \cdot e^{c_3 x_{11}} \cdot \log(c_4 x_2 + c_5)}{c_6 x_0 + c_7} + c_8 x_1 x_9 + c_9 x_{10} x_3 + c_{10} x_8 + c_{11} \right) \right) \quad (289)$$

Optimizing its constant with CMA-ES gives:

$$-0.0997 \sin \left(1.124 \sin \left(0.0750 x_1 x_9 + 0.06619 x_{10} x_3 + 0.3753 x_8 + 3.2948 + \frac{2.3662 e^{-0.6165 x_{11}} \log(1.0000 - 5.6576 \cdot 10^{-9} x_2)}{-1.9512 \cdot 10^{-8} x_0 - 2.0206 \cdot 10^{-7}} \right) \right) \quad (290)$$

With a performance in a cross-validation of $R^2 = 0.6499 \pm 0.0255$. While not bad, this is far from the $R^2 = 0.8148$ of the original SR's performance.

E.19 Task 361252, default hyperparameters

On task 361252 (video_transcoding), SR performs significantly worse than the other two algorithms. More importantly, its performance with enhanced hyperparameters is *worse* than with default hyperparameters, likely indicating issues with an incomplete exploration of the search space. The equations found by SR are:

$$y = \frac{x_{16} (x_{13} x_{13} - 0.299) 0.00328 - 0.442}{\frac{x_{13} x_{16}}{x_{15} (x_{14} \cdot 0.00328 + x_3)} + x_{13} - 0.963} - 0.211 \quad (291)$$

$$y = \frac{x_{16} \cdot 0.00297 x_{13} x_{13} - 0.582}{x_{13} - 0.963 + x_{16} \frac{1}{x_{14}} \cdot 11.7 + \frac{x_3 (-x_8 + \frac{x_9}{x_{15}})}{x_{11}}} \quad (292)$$

$$y = x_{16} \left(-x_{13} + \frac{0.715}{x_{13} - 1.17} \right) \left(x_{15} - \frac{x_{16}}{\frac{1}{x_3} (x_{14} - -0.0512)} - x_5 (-0.448) \right) (-0.000197) - 0.0565 \quad (293)$$

$$y = \left(x_{13} (x_{17} - x_5) - \left(x_{17} (-0.163) \frac{1}{x_{13} - 2.96} - -133. \right) \right) (x_{14} \cdot 1.79 \cdot 10^{-7} - -0.652) 0.00749 \frac{1}{x_{13} - 0.881 - 0.0274} \quad (294)$$

$$y = x_{13} (x_{13} - 0.574) - \frac{x_{16}}{x_{13} x_{15} \cdot 0.508 - 116. - \frac{1.97}{0.977 - (x_{13} + \frac{x_{16}}{x_{14}})}} - 6.23 \quad (295)$$

$$y = \frac{x_{13} x_{13} x_{13} (x_{17} \cdot 0.00237 - 0.449)}{x_{13} - 0.976 + x_{17} \cdot 7.92 \frac{1}{x_{14}}} + x_3 \cdot 0.00790 - 1.81 \quad (296)$$

$$y = x_{16} (-x_{15} - 0.813) \left(x_{13} (-0.000191) + \frac{8.75 \cdot 10^{-5} - \frac{3.85}{x_{14}}}{x_{13} - 1.06} \right) \quad (297)$$

$$y = \left(-\frac{x_{11}}{x_{12}(x_{13} - 1.21)} + x_{13} \right) 0.792 \left(\frac{x_{16}}{208. - \left(\frac{x_{15}}{0.350} + x_3 \cdot 0.0437 + \frac{x_{12}}{x_{12}} \right)} - 0.995 \right) \quad (298)$$

$$y = x_{16} (-3.32 \cdot 10^{-6}) \frac{1}{x_{13} - 3.10} \left(\frac{x_{17} - \frac{x_{16} + x_{17}}{x_{15}}}{x_{13} - 0.973 + x_{16} \cdot 5.83 \frac{1}{x_{14}}} - 0.140 \right) + 3.56 \quad (299)$$

$$y = x_{17} (x_{14} \cdot 3.96 \cdot 10^{-9} + 0.0290) + \frac{-0.316 + \frac{x_{14}(-8.81 \cdot 10^{-7}) + x_{17}(-0.0130) + 2.66}{x_{13} - 1.67}}{x_{13} - 0.276} - 4.83 \quad (300)$$

The equation found by the LLM is:

$$\hat{y} \approx c_0 + \frac{c_1 x_{16} x_{13}^{c_2} - c_3}{x_{13} - c_4 + c_5 \frac{x_{16}}{x_{14}}} \quad (301)$$

With a constant c_2 used to determine the power of a term $x_1 x_{16} x_{13}^{c_2}$.

Optimizing its constants with CMA-ES gives:

$$\hat{y} = \frac{3.0207 \cdot 10^{-4} x_{13}^{4.1056} x_{16} - 0.0675}{x_{13} - 0.9947 + \frac{1.2066 x_{16}}{x_{14}}} + 1.4197, \quad R2 = 0.7871 \quad (302)$$

And in a cross-validation $R2 = 0.7872 \pm 0.0113$. However, having a floating point power is not something which was possible using PySR's settings; so, we empirically set $c_2 = 3$ and rerun constant optimization with CMA-ES:

$$\hat{y} = \frac{9.825 \cdot 10^{-4} x_{13}^3 x_{16} - 0.2125}{x_{13} - 0.9828 + \frac{3.8670 x_{16}}{x_{14}}} + 0.9912, \quad R2 = 0.7823 \quad (303)$$

With a mean performance in a cross-validation of $R2 = 0.7823 \pm 0.0112$, so we accept the negligible decrease in performance to obtain a simpler expression.

E.20 Task 361253, enhanced hyperparameters

In task 361253 (wave_energy) with enhanced hyperparameters, no algorithm performs significantly better or worse than the others. The equations discovered by SR during the 10-fold cross-validation are:

$$y = -x_{27} + x_{31} + x_{32} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - (x_{12} - 2.39 \cdot 10) \quad (304)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} + \sin(x_2(-36.3)) - \sin\left(\frac{x_{16}}{x_{14}}\right) \quad (305)$$

$$y = x_{15} + x_{28} + x_{32} + x_{33} + x_{34} + x_{35} + x_{36}(-0.00575) + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{45} + x_{46} + x_{47} - \frac{x_{16}}{x_{14}} \quad (306)$$

$$y = x_{12} + x_{13} + x_{23} + x_{34} + x_{38} + x_{45} + x_{47} - (x_{16} - x_{19}) - (x_{21} - x_7) - (-x_{32} + (x_{28} - (x_{19} - -648.))170.) + (x_{36} + x_{46} - \frac{x_{16}}{x_{14}}) \quad (307)$$

$$y = -x_1 + x_{32} + x_{32} - x_{33} + x_{33} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - (x_0 - (x_{18} - \frac{x_{16}}{x_{14}})) \quad (308)$$

$$y = \left(x_{14} + x_{21} + x_{32} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{40} + x_{44} + x_{46} + \frac{x_{47}}{1.98} + x_{47} + \left(x_{21} + x_{21} + x_{32} + x_{32} + x_{33} + x_{36} + x_{36} + x_{37} + x_{37} + x_{38} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} \right) \right) \quad (309)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} \quad (310)$$

$$y = x_0 + x_{15} + x_{16} + x_{16} + x_2 + x_{20} - x_{26} + x_{27} - x_{29} - x_{30} + x_{33} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} \quad (311)$$

$$y = x_{21} + x_{21} + x_{21} + x_{21} + x_{23} + x_{23} - x_{25} + x_{26} + x_3 + x_{40} + x_{43} - (x_{19} + x_{41}) (-0.277) + (x_3 + x_3) 50.3 + \frac{x_{33} + x_{33} + x_{34} + x_{34} + x_{35} + x_{35} + x_{36} + x_{36} + x_{37} + x_{37} + x_{38} + x_{38} + x_{39} + x_{39} + x_{40} + x_{40} + x_{41} + x_{41} + x_{42} + x_{42} + x_{43} + x_{43} + x_{44} + x_{44} + x_{45} + x_{45} + x_{46} + x_{46} + x_{47} + x_{47}}{1.98} \quad (312)$$

$$y = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - (-x_{40} - 0.140) \quad (313)$$

The equations are quite surprising, as they are mostly linear compositions of the features. The equation proposed by the LLM-produced text is:

$$\hat{y} \approx c_0 + c_1 x_{32} + c_2 x_{33} + c_3 x_{34} + c_4 x_{35} + c_5 x_{36} + c_6 x_{37} + c_7 x_{38} + c_8 x_{39} + c_9 x_{40} + c_{10} x_{41} + c_{11} x_{42} + c_{12} x_{43} + c_{13} x_{44} + c_{14} x_{45} + c_{15} x_{46} + c_{16} x_{47} \quad (314)$$

which basically corresponds to a linear regression.

Optimizing its constants with CMA-ES gives:

$$\hat{y} = x_{32} + x_{33} + x_{34} + x_{35} + x_{36} + x_{37} + x_{38} + x_{39} + x_{40} + x_{41} + x_{42} + x_{43} + x_{44} + x_{45} + x_{46} + x_{47} - 105.301, \quad R^2 = 1.0 \quad (315)$$

It's interesting to note that the coefficients $c_1, \dots, c_{11}$ are not all exactly set at 1.0, but can have values like $c_1 = 0.99995864816671$ or $c_2 = 1.00000305084385$. Optimizing the constants in a cross-validation gives $R^2 = 1.0000 \pm 0.0000$, which is of course an approximation, but still indicates an excellent fitting.

E.21 Task 361266, enhanced hyperparameters

On task 361266 (kings_county) SR performs significantly worse than the other algorithms. The equations discovered by SR during the cross-validation are:

$$y = x_1 + x_{14} \left(1.48x_{12} + 1.48x_{16} + 1.48x_2 + 24.6e^{x_7} + 20.2 + \frac{26.5 \left(-x_{19} \log(x_2) + \frac{x_2(x_6 - 2.77)}{x_8} \right)}{x_8} \right) + 4.18x_9 - 4.18(x_{19} + 1.14x_{14}) \quad (316)$$

$$y = \left(x_8 - \left(x_{18} - \left(x_5 - \cos \left(x_{14} - \left(x_8 + \frac{1.99}{x_{16}(-x_{18} - 0.678 + 1.40)} \right) \right) \right) \right) - \left(x_5 - \cos \left(x_{14} - \sin \left(\frac{x_{15}}{x_{16}(0.725 - x_{18}) + x_8} \right) \right) \right) \right) \quad (317)$$

$$y = 0.725x_{12}x_7(1.59x_1x_8 - 1.59) - x_2x_5 \left(-x_8 - 12.2 - \frac{1.97}{x_8 \sin(x_{14})} \right) \log(x_3) - x_8 + \left(-x_{15} + \left(\frac{x_2}{x_3} + x_8 \right) (6.78x_7 + x_8 + 2.1) \right) \quad (318)$$

$$y = (x_5 + 1.15) \left(x_8 (-x_{12} (-x_0 + \cos(x_{14}))) + x_{12} + x_{16} \right) + \left(\frac{x_{10}}{(-0.287) 1.11} + x_{13} x_{13} + x_{13} + x_{16} + x_{16} + x_{18} + (-(-(x_1 + 1. \right. \quad (319)$$

$$y = x_8 \left(272. - \frac{1.27 \cdot 10^4}{x_{14}} \right) \left(-(-2.28 \cdot 10^4) \left(\frac{x_{14}}{x_2} + x_5 \right) + \left(x_{12} + x_2 \left(\sin \left(-\frac{2.30 \cdot 10^4}{x_{14}} \right) + 2.01 \right) + x_6 + 2.28 \cdot 10^3 \right) 1.73 \right) \quad (320)$$

$$y = x_{12} + x_4 (x_0 + x_5 (-0.560)) \left(\frac{x_{12} x_6 x_8}{x_5 + \frac{x_{13} - 1.02}{2.31}} + x_{12} \right) + x_6 x_8 (x_{16} + x_{16} - -1.85) - \left(-x_7 + \left(x_{12} (-4.79) + x_{13} x_7 - x_{16} - \right. \quad (321)$$

$$y = -x_{14} - 239.x_2 - x_8 (x_2 (-44.9) + x_7) + x_8 (x_{14} + e^{0.520} - -10.7) 15.6 (e^{x_6} - 1.16) + (x_{14} + x_8 (-22.0)) ((x_{15} - (x_{12} - (\quad (322)$$

$$y = -x_2 + \frac{x_1 \left(x_0 (-x_{19} + x_9 - 1.32 \cdot 10^3) + \frac{x_{16} x_7 \cos \left(\frac{x_{14} (-19.8)}{0.319} \right) + x_7 - 1.98 \right)}{0.633} - x_{12} - x_{20} + (x_{14} \cdot 17.8 - 834.) \left(-x_0 - x_{11} + \left(x_0 x_8 - \left(-x_{10} \right. \right) \quad (323)$$

$$y = -1.95x_{11} - 3.95x_9 + 27.5 (-x_1 + x_{11} + x_2) e^{x_5 - \sin(6.76e^{-\sin(x_{14})})} + (x_7 + x_8 + 14.7) \left(2x_{11} + x_{18} + x_2 (x_6 - 15.2) + x_2 (x \quad (324)$$

$$y = x_{10} + \frac{x_{12}}{0.355} - x_9 + \left(x_{12} + x_8 \left(x_5 + \frac{x_6}{x_8 + 1.05} + 1.62 \right) \right) \left(\frac{\left(-1 \right) 1.33 x_2 \left(-0.0397 + \frac{1.62}{-x_2 + x_{20}} \right) \left(x_{14} \left(\cos \left(\frac{x_{14}}{(-1) 0.844 \frac{1}{x_8}} \right) \right) (- \right. \right) \quad (325)$$

The equation proposed in the text produced by the LLM is:

$$\hat{y} \approx c_0 + c_1 x_{12} + c_2 x_2 + c_3 x_6 x_8 + c_4 e^{c_5 x_6 + c_6 x_7} + c_7 \sin(c_8 x_{14} + c_9) + \quad (326)$$

$$+ c_{10} \log(c_{11} x_2) + \frac{c_{12} x_1 + c_{13}}{c_{14} x_8 + x_{13} - x_{14}} \quad (327)$$

Optimizing its constants with CMA-ES does not produce any meaningful result, as the optimization seems unable to find good solutions, probably due to the presence of log. After a human intervention, the equation now looks like:

$$\hat{y} \approx c_0 + c_1 x_1 2 + c_2 x_1 1 + (x_2 + c_3 x_5 + c_4 x_{14} + c_5) (c_6 x_6 + c_7 x_8 + c_9) + c_{10} \sin(c_{11} x_{14} + c_{12}) \quad (328)$$

And optimizing its constants with CMA-ES gives:

$$\hat{y} = 31.867x_{11} + 74.53x_{12} + (20.352x_6 + 44.071x_8 - 226.401) (42.249x_{14} + x_2 + 2397.756x_5 - 2121.226) + 3828.561 \sin(21.852x_{14} + c_{12}) \quad (329)$$

With a 10-fold cross-validation performance of $R2 = 0.6434 \pm 0.0431$.

Still, the R^2 value is lower than the original. Restarting from Eq. 320 found for fold 4:

$$\hat{y} = x_8 \left(272. - \frac{1.27 \cdot 10^4}{x_{14}} \right) \left(-(-2.28 \cdot 10^4) \left(\frac{x_{14}}{x_2} + x_5 \right) + \left(x_{12} + x_2 \left(\sin \left(-\frac{2.30 \cdot 10^4}{x_{14}} \right) + 2.01 \right) + x_6 + 2.28 \cdot 10^3 \right) 1.73 \right) \quad (330)$$

$$\hat{y} \approx x_8 \left(c_0 - \frac{c_1}{x_{14}} \right) \left(c_2 \left(\frac{x_{14}}{x_2} + x_5 \right) + x_{12} + c_3 x_2 \left(\sin \left(\frac{c_4}{x_{14}} \right) + x_6 + c_5 \right) \right) \quad (331)$$

Asking the LLM to simplify the equation gives:

$$\hat{y} \approx x_8 \left(1 - \frac{c_0}{x_{14}} \right) \left(c_1 \frac{x_{14}}{x_2} + c_2 x_5 + c_3 x_{12} + c_4 x_6 + x_2 \left(c_5 \sin \left(\frac{c_6}{x_{14}} \right) + c_7 \right) + c_8 \right) \quad (332)$$

However, all optimizations lead to inferior results.

E.22 Task 361268, enhanced hyperparameters

On task 361268 (fps_benchmark), SR performs significantly worse than the other two algorithms. The equations found by SR during the cross-validation are:

$$y = x_{29} + \left(\left(x_8 - \frac{\sin(x_{30} + x_{36} - \sin(x_{36} - (x_{30} - \sin(x_{30} + x_{38}) + 0.0410) - 0.811))}{0.0498} - 0.855 \right) 1.19 - -0.859 \right) 1.75 + \frac{-(x_{30} + x_{36} - \sin(x_{36} - (x_{30} - \sin(x_{30} + x_{38}) + 0.0410) - 0.811))}{0.0498} \quad (333)$$

$$y = 4.09x_1 + x_{10} + 3.54x_{20} + 2x_{24} - x_{25} + x_{29} + 2x_{30} - 3.54x_{35} + x_{38}(-1.61x_{38} - 7.46) + \log(x_6) \cos(0.0961x_{34}) + 2.54 \log(x_6) \quad (334)$$

$$y = 0.846x_{18} + 20.2x_{25} + 1.95x_{30} + 0.343x_{32} - 0.846x_{36} - 6.73x_{38} + 0.585x_9(x_{14} - 1.80) + 6.73x_9 + 6.73 \log(x_1 + x_{17}) + 1.86 \log(x_1 + x_{17}) \quad (335)$$

$$y = 0.765x_{11} + 2x_{20} + 0.765x_{29} - 15.7x_{38} + 41.6e^{\sin(\sin(-2x_{36} + \log(x_{36} + 0.0481) + 1.25))} - 5.29 + \frac{7.67e^{-x_{36}}}{x_{36} + 0.0526} + \frac{0.784}{x_{27}} \quad (336)$$

$$y = -0.539x_{14} + x_{19} \left(-2.13x_{35} - 2.13x_{36} \sin(x_{36}) + \cos(x_{26}) + 15.8 + \frac{12.1}{x_{38}} \right) + x_{29} - (14.7 - \log(x_{38})) \sin(x_{36}) + (x_{26} - x_{38}) \cos(x_{36}) \quad (337)$$

$$y = 1.74x_1 - 1.74x_{18} + 7.74x_{24} + x_{29} + 1.74x_3 - 1.74e^{x_{35}} - 1.74e^{x_{38}} + 1.74e^{\cos(x_{36} + 1.87)} \log(0.580x_{34}) - 44.4e^{\cos \left(e^{-x_{36} + \frac{e \cos(x_{36})}{\cos(x_{36})}} \right)} \quad (338)$$

$$y = x_{11} - x_{19} + x_{29} \cos(\cos(x_{38})) + 2.52x_{30} - 3x_{38} - 2.52x_9 + 5.97(2.02 - 33.0 \sin(x_{36} - 0.727))e^{-x_{36}} + (x_{29} + (\sin(x_{36} - 0.727))e^{-x_{36}}) \quad (339)$$

$$y = x_1 \left(0.930 + \frac{x_{30} + 1.06}{x_{13}} \right) + 0.215x_{11} + 2x_{24} - 0.430x_{28}x_{38} + 0.859x_{29} + 0.430x_3 - 0.215x_{36}x_{38} - 0.215x_{38}^2 - 0.215x_{38} + 0 \quad (340)$$

$$y = 1.76x_1 + x_{14} + x_{18} + x_{19} + x_{20}^2 + 1.76x_{20} + x_{24} + 1.76x_3 - 0.766x_{36} - 2x_{38} - 1.76x_8 \cos(2.09 \sin(2.67 \cos(0.472x_{36} - 0.95x_{38}))) \quad (341)$$

$$y = \frac{x_1}{x_{36} - 12.8 - 0.0471} + \frac{x_1}{x_{36} - 0.0471} + x_{14} - x_{18} + x_{20}x_{38} + x_{20} - x_{28} - x_{29}(x_{19} - 0.964) + x_3 + x_{38} + x_9 + x_9 + x_9. \quad (342)$$

The equation proposed by the LLM is:

$$\hat{y} \approx c_0 + \sum_{i=1}^k c_i x_i + c_{k+1} \cos(c_{k+2} \sin(c_{k+3} x_{36} + c_{k+4})) + \frac{c_{k+5}}{c_{k+6} - x_{36}} + \frac{c_{k+7}}{x_{38}} \quad (343)$$

With the most represented variables being $x_1, x_{29}, x_{30}, x_{20}$. After several iterations of human intervention, the equation now looks like:

$$\hat{y} \approx c_0 + c_1 x_1 + c_2 x_{20} + c_3 x_{29} + c_4 x_{30} + \frac{c_5}{x_{36} - c_6} + \frac{c_7}{x_{36} - c_8} + c_9 x_{38}^2 + c_{15} x_{38} x_{30} + \frac{c_{10}}{x_{38}} + c_{11} \cos(c_{12} \sin(c_{13} x_{36} + c_{14})) \quad (344)$$

Optimizing its constants with CMA-ES gives:

$$\hat{y} = 3.431x_1 + 2.782x_{20} + 0.873x_{29} + 0.333x_{30}x_{38} + 1.946x_{30} - 35.873x_{38}^2 + 39.939 \cos(25.355 \sin(6.623x_{36} - 3.141)) + 444.305 + \frac{3}{x_{36}} \quad (345)$$

And a performance in a 10-fold cross-validation of $R^2 = 0.7066 \pm 0.0099$.

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